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STATISTICAL REVIEW(S)



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Center for Drug Evaluation and Research
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STATISTICAL REVIEW AND EVALUATION

CLINICAL STUDIES

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Indication(s): [REDACTED] (b) (4)
Applicant: Novartis
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Keywords: active control, Cox regression, combination products

Table of Contents

EXECUTIVE SUMMARY	5
INTRODUCTION	6
1.1 OVERVIEW.....	6
1.2 DATA SOURCES	6
STATISTICAL EVALUATION	7
1.3 DATA AND ANALYSIS QUALITY	7
1.4 EVALUATION OF EFFICACY	7
1.4.1 <i>Study Design and Endpoints</i>	7
1.4.2 <i>Statistical Methodologies</i>	8
1.4.3 <i>Patient Disposition, Demographic and Baseline Characteristics</i>	9
1.4.4 <i>Results and Conclusions</i>	11
1.5 EVALUATION OF SAFETY	17
1.6 BENEFIT-RISK ASSESSMENT (OPTIONAL).....	17
FINDINGS IN SPECIAL/SUBGROUP POPULATIONS	17
1.7 GENDER, RACE, AGE, AND GEOGRAPHIC REGION	17
1.8 OTHER SPECIAL/SUBGROUP POPULATIONS	18
SUMMARY AND CONCLUSIONS	19
1.9 STATISTICAL ISSUES	19
1.10 COLLECTIVE EVIDENCE	19
1.11 CONCLUSIONS AND RECOMMENDATIONS	19
1.12 LABELING RECOMMENDATIONS (AS APPLICABLE).....	20

LIST OF TABLES

Table 1 List of all studies included in analysis	6
Table 2 Reasons for treatment discontinuation.....	10
Table 3 Patient demographic and baseline characteristics.....	11
Table 4 Adjudicated Primary Endpoint Analysis (CV death or HF hospitalization).....	12

LIST OF FIGURES

Figure 1 Weights for alpha relocation in the sequentially rejective multiple test procedure for the secondary hypotheses	9
Figure 2 Proportion of subjects included in primary analysis over time in the study.....	13
Figure 3 Kaplan-Meier estimates of event rates over time.	14
Figure 4 Log{-log(survival)) plot for primary endpoint (Time is shown on the log-scale on the x-axis, log(-log(estimated survival)) on the y-axis).	15
Figure 5 Kaplan-Meier estimates of all-cause mortality rates (secondary endpoint).....	16
Figure 6 High dose/warfarin results for subgroups defined by gender, race, age, and geographic region.	18

EXECUTIVE SUMMARY

There was only one phase 3 trial for this indication in the submission. The study drug is a combination of valsartan and sacubitril. Both valsartan (a component of the combination study drug) and enalapril (the active control) are approved for the treatment of heart failure, but there has never been a study comparing valsartan monotherapy to enalapril. According to their respective labels, enalapril has a mortality effect compared to placebo, but valsartan has no mortality effect compared to placebo (where subjects were on the background therapy recommended by their physicians and 93% of the subjects were on ACE-inhibitor therapy, of which enalapril is one such therapy). In the trial, the combination was effective compared to the control on two endpoints: time to first heart failure hospitalization or CV death (primary endpoint) and time to all-cause mortality (a secondary endpoint). There is no way to know from this trial whether both components in the combination contribute to both or either of the claimed effects. From the valsartan label, it may be reasonable to infer that the valsartan component does not contribute to the claimed mortality effect; valsartan had no effect compared to placebo in the Val-HeFT trial when ACE-inhibitors were a background therapy, so it is not likely that valsartan would be more effective than an ACE-inhibitor in a head-to-head comparison. Although valsartan alone reduced HF hospitalization compared to placebo in the Val-HeFT trial, it is unknown whether both components contribute to the combined effect of valsartan/sacubitril compared to enalapril on the composite primary endpoint in this trial.

INTRODUCTION

1.1 Overview

Table 1 List of all studies included in analysis

	Phase and Design	Treatment Period	Follow-up Period	# of Subjects per Arm	Study Population
<i>Study CLCZ696B2314</i>	<i>Phase 3</i>	<i>Up to 51 months (median of 27 months)</i>	<i>Up to 51 months (median of 27 months)</i>	<i>4209 (LCZ696), 4233 (enalapril)</i>	<i>patients with CHF (NYHA class II - IV), age 18 years or older, LVEF $\leq$ 35%, B-type natriuretic peptide (BNP) $\geq$ 150 pg/ml (or N-terminal prohormone B-type natriuretic peptide [NT-proBNP] $\geq$ 600 pg/ml) or BNP $\geq$ 100 pg/mL (or NT-proBNP $\geq$ 400 pg/ml) and a hospitalization for HF within the last 12 months.</i>

Source: Study Report.

1.2 Data Sources

Electronic datasets and Study Reports:

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<\\cdsesub1\evsprod\NDA207620\0002\m5\datasets\lcz696b2314\analysis\legacy\datasets>

STATISTICAL EVALUATION

1.3 Data and Analysis Quality

Both the data quality or analysis quality were very good to excellent. The largest file in the submission was about 640 MB. This file, although large, was not too large for me to open on my workstation. In the data analysis, appropriate adjustments were made for multiple comparisons of different endpoints and sequential testing of these endpoints at different time points (interim monitoring).

1.4 Evaluation of Efficacy

1.4.1 Study Design and Endpoints

This was an active controlled trial without placebo. Although the active control has an effect described in the label, the trial was designed to show superiority to the active control and there was no non-inferiority testing planned. Unblinded ACEI or ARB concomitant therapy was not permitted in the trial.

The study population consisted of patients with CHF (NYHA class II - IV), aged 18 years or older with LVEF $\leq 40\%$, changed to $\leq 35\%$ by Protocol Amendment 1. Patients eligible for inclusion in the study had to have B-type natriuretic peptide (BNP) ≥ 150 pg/ml (or N-terminal prohormone B-type natriuretic peptide [NT-proBNP] ≥ 600 pg/ml) or BNP ≥ 100 pg/mL (or NT-proBNP ≥ 400 pg/ml) and a hospitalization for HF within the last 12 months. Eligible patients were to have been on an angiotensin converting enzyme inhibitor (ACEI) or an angiotensin receptor blocker (ARB), at a stable dose of at least 10 mg/day of enalapril or equivalent for at least 4 weeks prior to Visit 1. Patients were to have been treated with a β -blocker, unless contraindicated or not tolerated, at a stable dose for at least 4 weeks prior to Visit 1. An aldosterone antagonist was also to have been considered in all patients if indicated.

A total of 8442 patients were randomized (4209 patients to LCZ696 and 4233 patients to enalapril). Forty-three patients were excluded from the efficacy analysis due to site closure for serious GCP violations (37 patients) or due to misrandomization (6 patients). Thus, the full analysis set (FAS) included 8399 patients (4187 patients in the LCZ696 group and 4212 patients in the enalapril group).

The primary endpoint was the time to first heart failure hospitalization or CV death. There were 4 secondary endpoints: i) time to all-cause death; ii) change from baseline (CFB) (randomization visit) in the clinical summary score for HF symptoms and physical limitations (as assessed by KCCQ) at 8 months; iii) time to new onset of atrial fibrillation; iv) time to first occurrence of either (1) a 50% decline in eGFR relative to baseline, (2) >30 mL/min/1.73 m² decline in eGFR relative to baseline to a value below 60 mL/min/1.73 m², or (3) reaching ESRD.

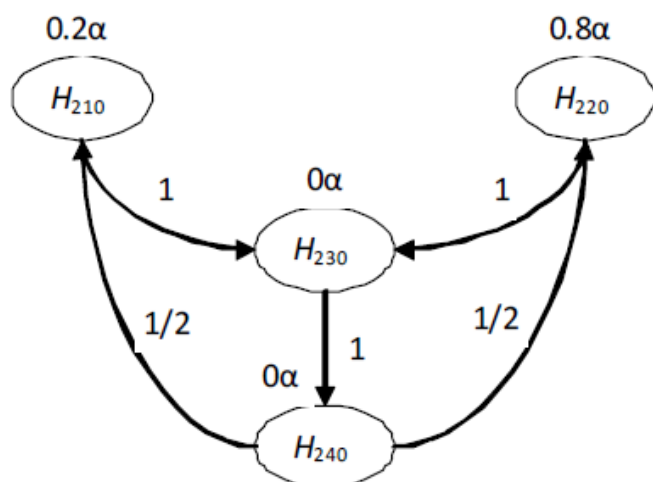
1.4.2 Statistical Methodologies

For the primary composite endpoint and the three time-to event secondary endpoints, time to event was analyzed using a Cox regression model with terms for treatment and region. For the remaining secondary endpoint- change in KCCQ score- a repeated measures ANCOVA analysis was used with terms for treatment, region, and time point (Months 4 and 8).

The study was designed to have approximately 2410 primary endpoint events and 1229 CV-death events if the study reached the maximum study duration. Three interim analyses were planned at 1/3, 1/2 and 2/3 of information time (i.e. approximately 804, 1205 and 1607 patients, respectively, with a primary event). However, some adjustment to the frequency and time for interim analyses was made to coincide with the DMC meetings. Appropriate statistical adjustments for the interim analyses actually performed were made to control the overall type-I error of 0.025 (one-sided). For each IA, the analysis dataset comprised all patients who were randomized before the cutoff date. The Haybittle-Peto type of boundary was used for the interim efficacy analyses to assess superiority. The interim efficacy analysis with the boundary spent approximately an alpha of 0.0001 (one-sided) at the first IA and 0.001 (one-sided and nominal) at the second and third interim analyses.

Whatever time point the study stopped, if the primary endpoint was statistically significant, then the secondary endpoints were to be tested using the same alpha as used for the primary endpoint. The testing of the four secondary endpoints was to be done using the Bonferroni-Holm's method as shown in the graph in Figure 1.

Figure 1 Weights for alpha relocation in the sequentially rejective multiple test procedure for the secondary hypotheses



H_{210} denotes the null hypothesis for the comparison of KCCQ clinical summary score.

H_{220} denotes the null hypothesis for the comparison of all-cause mortality.

H_{230} denotes the null hypothesis for the comparison of new onset AF.

H_{240} denotes the null hypothesis for the composite renal endpoint.

Source: Figure 9-2 of Study Report.

1.4.3 Patient Disposition, Demographic and Baseline Characteristics

The reasons for treatment discontinuation are shown in **Table 2** and the baseline and demographic characteristics are shown in Table 3. The first table shows that very few subjects were lost to follow-up except for those who died. However, about 30% of the subjects stopped double-blind treatment before the end of the study. A strength of the trial is that there is very little loss to follow-up and subjects were followed after stopping study drug. In Table 3, it is seen that the average age was about 64 years, 78% were male, 66% were Caucasian. There were no significant differences in the demographics between groups.

Table 2 Reasons for treatment discontinuation.

	LCZ696 N=4209 n (%)	Enalapril N=4233 n (%)	Total N=8442 n (%)
Patients who never received study treatment during double-blind period	6 (0.14)	4 (0.09)	10 (0.12)
Patients who permanently discontinued study treatment during double-blind period	1182 (28.08)	1353 (31.96)	2535 (30.03)
Primary reason for EOS treatment			
AE(s)	437 (10.38)	510 (12.05)	947 (11.22)
Abnormal laboratory value(s)	7 (0.17)	6 (0.14)	13 (0.15)
Unsatisfactory therapeutic effect	0 (0.00)	1 (0.02)	1 (0.01)
Patient's condition no longer requires study drug	3 (0.07)	4 (0.09)	7 (0.08)
Lost to follow-up	8 (0.19)	8 (0.19)	16 (0.19)
Administrative problems	12 (0.29)	9 (0.21)	21 (0.25)
Death	430 (10.22)	512 (12.10)	942 (11.16)
Protocol deviation	5 (0.12)	3 (0.07)	8 (0.09)
Patient's request	208 (4.94)	219 (5.17)	427 (5.06)
Other	72 (1.71)	81 (1.91)	153 (1.81)
Missing end of treatment	5 (0.12)	3 (0.07)	8 (0.09)
Patients completed study treatment*	3022 (71.80)	2877 (67.97)	5899 (69.88)

Source: [Table 14.1-1.3](#)

There were 10 patients in the Randomized set who did not receive study treatment, including 6 patients who were randomized erroneously (misrandomized).

*Patients still on treatment at study close-out.

Source: Table 10-3 of Study Report.

Table 3 Patient demographic and baseline characteristics

Variable/ Statistic/category	LCZ696 N=4209	Enalapril N=4233	Total N=8442
Age (years)			
n	4209	4233	8442
Mean	63.78	63.82	63.80
SD	11.520	11.250	11.385
Q1	57.00	57.00	57.00
Median	64.00	64.00	64.00
Q3	72.00	72.00	72.00
Age category - n (%)			
< 65 years	2122 (50.42)	2177 (51.43)	4299 (50.92)
>= 65 years	2087 (49.58)	2056 (48.57)	4143 (49.08)
< 75 years	3423 (81.33)	3450 (81.50)	6873 (81.41)
>= 75 years	786 (18.67)	783 (18.50)	1569 (18.59)
Gender - n (%)			
Male	3321 (78.90)	3274 (77.34)	6595 (78.12)
Female	888 (21.10)	959 (22.66)	1847 (21.88)
Predominant race - n (%)			
Caucasian	2780 (66.05)	2799 (66.12)	5579 (66.09)
Black	213 (5.06)	215 (5.08)	428 (5.07)
Asian	760 (18.06)	750 (17.72)	1510 (17.89)
Native American	84 (2.00)	88 (2.08)	172 (2.04)
Pacific Islander	0 (0.00)	1 (0.02)	1 (0.01)
Other	372 (8.84)	380 (8.98)	752 (8.91)
Ethnicity - n (%)			
Hispanic/Latino	777 (18.46)	778 (18.38)	1555 (18.42)
Chinese	242 (5.75)	251 (5.93)	493 (5.84)
Indian	335 (7.96)	333 (7.87)	668 (7.91)
Japanese	2 (0.05)	0 (0.00)	2 (0.02)
Mixed ethnicity	32 (0.76)	39 (0.92)	71 (0.84)
Other	2821 (67.02)	2832 (66.90)	5653 (66.96)
Region(1)			
North America	310 (7.37)	292 (6.90)	602 (7.13)
Latin America	726 (17.25)	732 (17.29)	1458 (17.27)
Western Europe	1029 (24.45)	1028 (24.29)	2057 (24.37)
Central Europe	1398 (33.21)	1439 (33.99)	2837 (33.61)
Asia/Pacific and other	746 (17.72)	742 (17.53)	1488 (17.63)

Source: Table 11-3 of Study Report.

1.4.4 Results and Conclusions

The trial was stopped at the third interim analysis by the DMC because the efficacy boundary was crossed. The combination was superior to the control on the primary endpoint. The results are shown in Table 4. The p-value was less than 0.0001, which was statistically significant and smaller than the boundary pre-specified by the Haybittle-Peto rule (0.001). The first composite outcome event was CV death in 377 patients and 459 patients in the LCZ696 group and the

enalapril group, respectively. This can be found by subtracting the number of patients where the first event was HF hospitalization from the total number of primary endpoint events.

Table 4 Adjudicated Primary Endpoint Analysis (CV death or HF hospitalization)

Response variable	LCZ696 n/N (%)	Enalapril n/N (%)	LCZ696 n/T (EAIR)(1) (95% CI)	Enalapril n/T (EAIR)(1) (95% CI)	HR (95% CI)
Primary Composite	914/4187 (21.83)	1117/4212 (26.52)	914/87.22 (10.48) (9.81,11.18)	1117/84.93 (13.15) (12.39,13.95)	0.80 (0.73,0.87)
CV death	558/4187 (13.33)	693/4212 (16.45)	558/93.08 (5.99) (5.51,6.51)	693/92.35 (7.50) (6.96,8.08)	0.80 (0.71,0.89)
1st HF Hospitalization	537/4187 (12.83)	658/4212 (15.62)	537/87.22 (6.16) (5.65,6.70)	658/84.93 (7.75) (7.17,8.36)	0.79 (0.71,0.89)

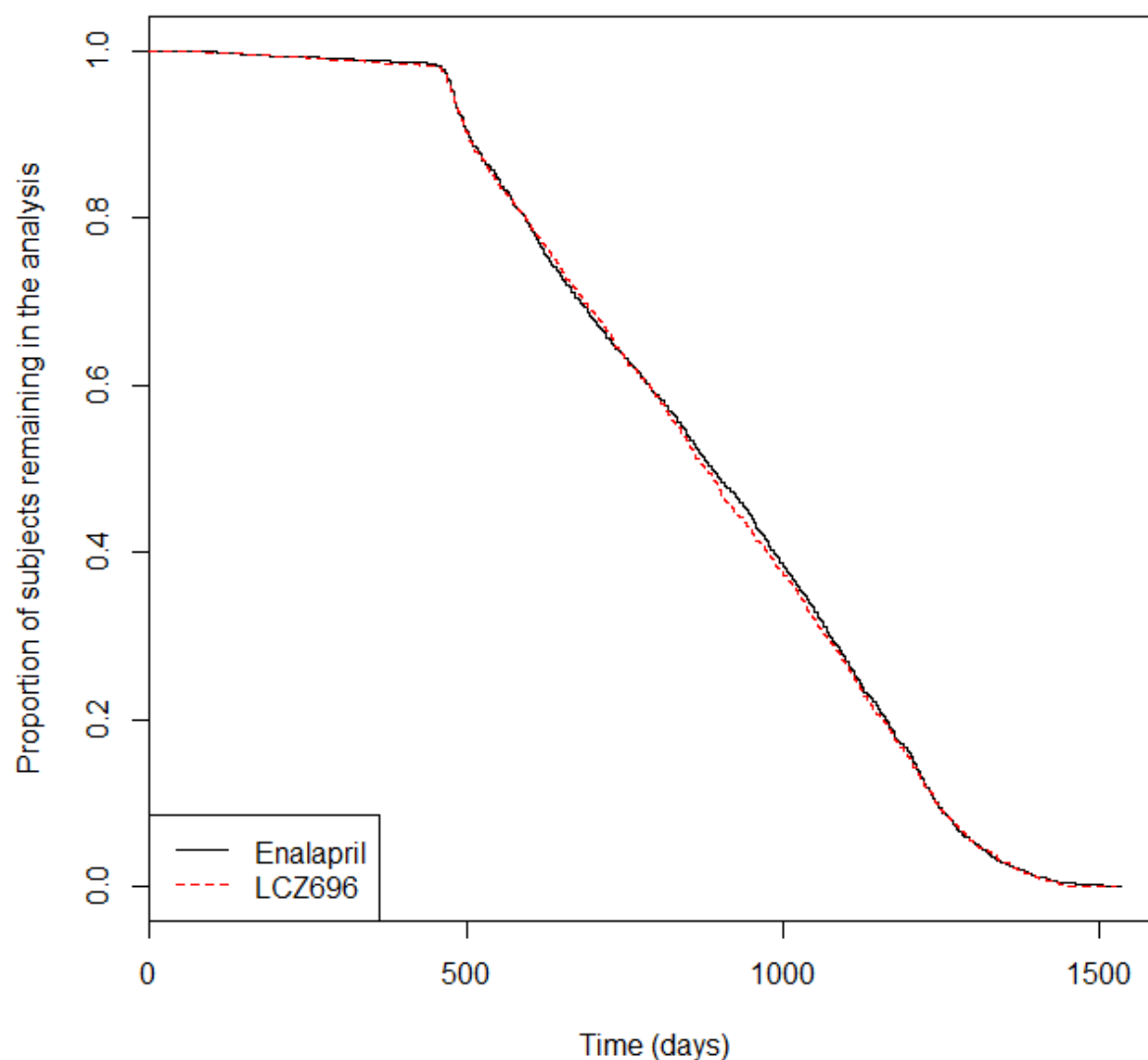
n: Total number of events included in the analysis; N: Total number of patients included in the analysis.

(1) EAIR (Exposure-adjusted incidence rate per 100 patient years)= n/T: T(100 years): total up-toevent/ censoring duration-time summarized over patients in the respective treatment group.

Source: Table 11-4 of Study Report and confirmed by FDA.

Figure 2 shows the estimated proportion of subjects remaining in the analysis over time. These are Kaplan-Meier type estimates using the reverse Kaplan-Meier method (Altman, D. G., et al. "Review of survival analyses published in cancer journals." *British Journal of Cancer* 72.2 (1995): 511.). There are two curves in the figure, but they are nearly identical. This figure is for the time to followup and not for the time on double-blind treatment.

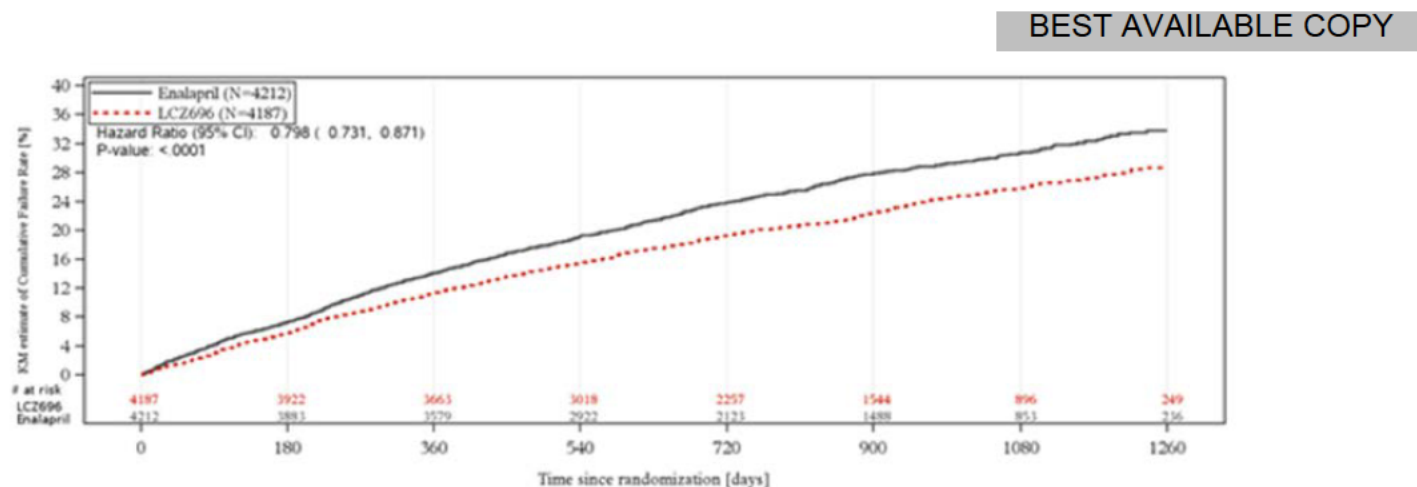
Figure 2 Proportion of subjects included in primary analysis over time in the study.



Source: FDA analysis.

Next, in Figure 3, the Kaplan-Meier estimates for the primary endpoint are shown. This shows the estimate of the proportion of people with a primary endpoint event.

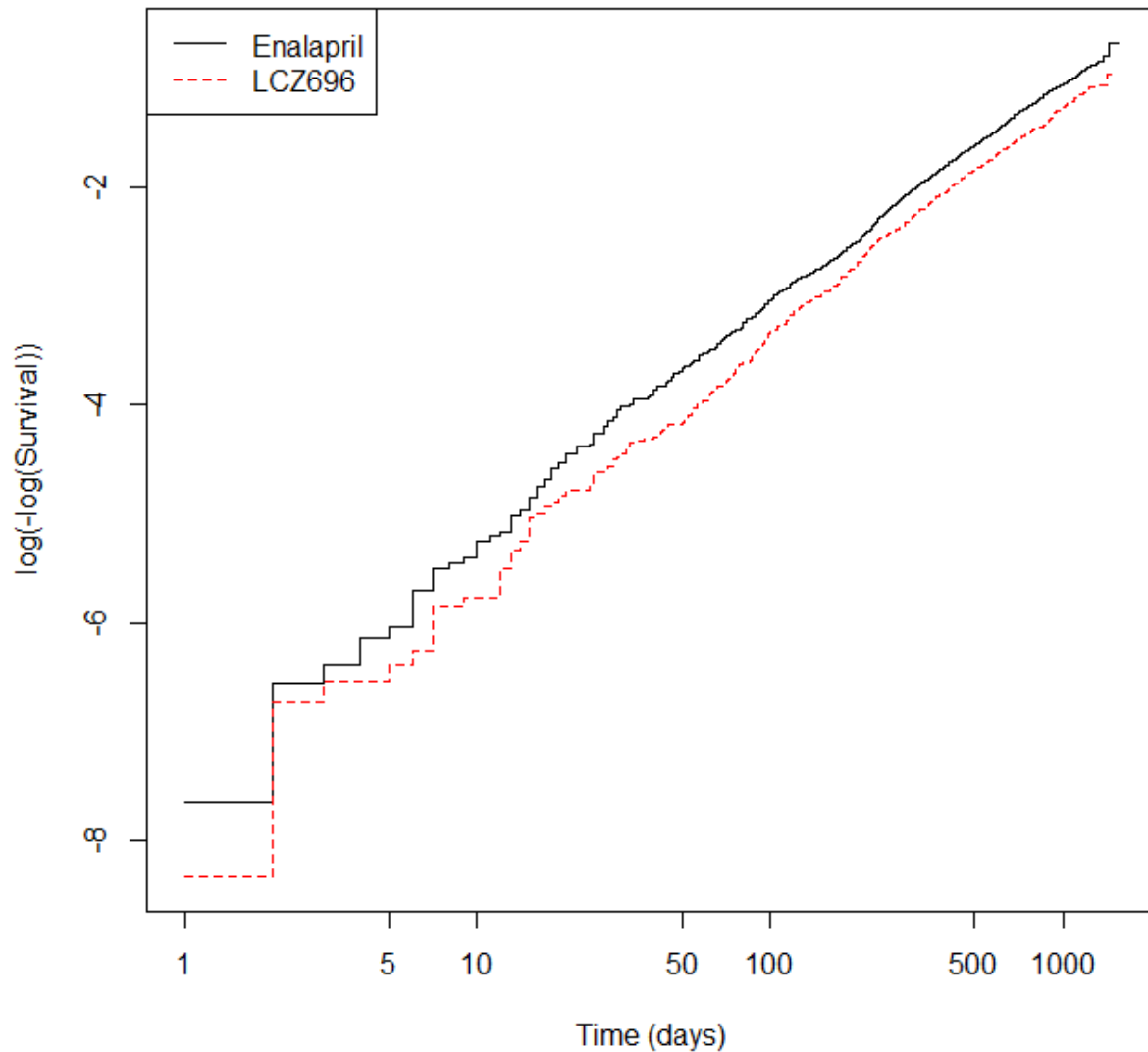
Figure 3 Kaplan-Meier estimates of event rates over time.



Source: Figure 11-1 of Study Report

In Figure 4, the complimentary log-log survival plots are shown. These two curves are approximately straight lines and approximately parallel. This suggests that the time to events can be approximately modelled by the parametric Weibull survival family and that the proportional hazards assumption is not clearly violated.

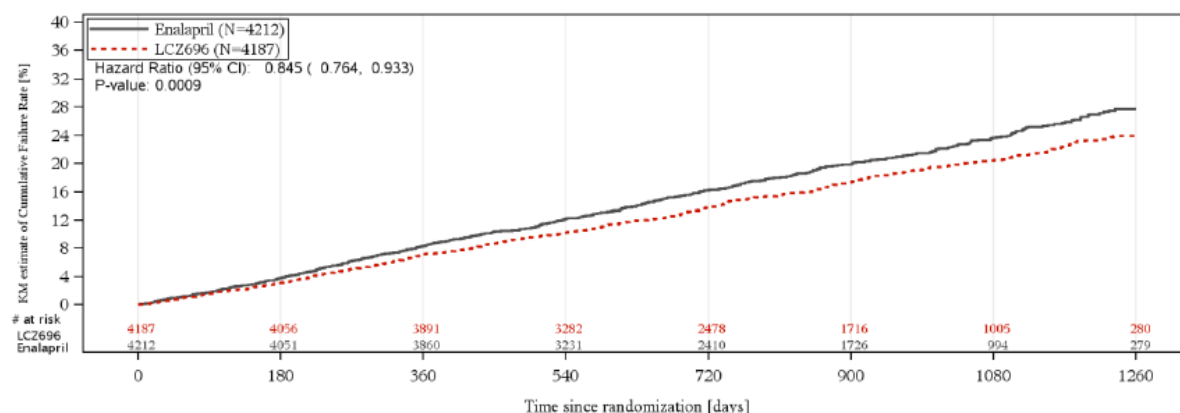
Figure 4 Log{-log(survival)) plot for primary endpoint (Time is shown on the log-scale on the x-axis, log(-log(estimated survival)) on the y-axis).



Source: FDA analysis

Of the 4 secondary endpoints, only all-cause mortality was statistically significant. The Kaplan-Meier curves are shown in Figure 5. The p-value in the figure is a two-sided p-value. This p-value is significant because the one-sided p-value was to be compared to 0.8×0.001 to determine whether it is significant (the one-sided p-value 0.0005 was smaller than 0.0008).

Figure 5 Kaplan-Meier estimates of all-cause mortality rates (secondary endpoint).



Source: [Figure 14.2-2.3](#)

The HR and its confidence intervals are estimations from a Cox-regression model with treatment and region as fixed factors. p-value is 2-sided and is based on this model.

Source: Study report Figure 11-4

The dose of valsartan in the LCZ696 combination is equivalent to 160 mg of valsartan administered bid. Valsartan is an ARB (angiotensin receptor blocker) approved with a heart failure indication in the label on the basis of the Val-HeFT trial. In that trial, valsartan was compared to placebo where everyone in the trial was encouraged to be on the background therapy as chosen by their physician (93% were on a background of ACE-inhibition therapy). In the overall population, there was no mortality effect, but a 13% reduction in heart failure morbidity (defined as all-cause mortality, sudden death with resuscitation, hospitalization for heart failure, and the need for intravenous inotropic or vasodilatory drugs for at least 4 hours) including a 27% reduction in heart failure hospitalization (decrease from 18% to 14%). Here, the combination was compared to enalapril 10 mg twice daily. Enalapril is approved for treatment of symptomatic ^{(b) (4)} heart failure. In the trial for symptomatic patient (SOLVD-Treatment), the use of enalapril was associated with an 11% reduction in all-cause mortality and a 30% reduction in hospitalization for heart failure compared to placebo. In that study, the dose of enalapril was titrated up to a maximum of 10 mg twice daily if the patient did not have symptomatic hypotension or worsening renal function. The FDA label recommends a maintenance dose of 2.5 to 20 mg daily in two divided doses with a maximum recommended dose of 40 mg daily (in two divided doses). In the only trial where 40 mg daily dose was studied, most patients did not tolerate this dose.

In the trial, the combination was effective compared to the control on two endpoints: time to first heart failure hospitalization or CV death (primary endpoint) and time to all-cause mortality (a secondary endpoint). There is no way to know from this trial whether both components in the combination contribute to both or either of the claimed effects. From the valsartan label, it may be reasonable to infer that the valsartan component does not contribute to the claimed mortality effect; valsartan had no effect compared to placebo in the Val-HeFT trial when ACE-inhibitors were a background therapy, so it is not likely that valsartan would be more effective than enalapril in a head-to-head comparison. Although valsartan alone reduced HF hospitalization compared to placebo in the Val-HeFT trial, it is unknown whether both components contribute to the combined effect of valsartan/sacubitril compared to enalapril on the composite primary endpoint in this trial. More thought should have been taken to design a trial that would show both components contribute to the claimed effect. Certainly, a design with three treatment arms (valsartan alone, sacubitril alone, and the combination) where everyone was on a background per the physician's judgment (as done in Val-HeFT) would have been ethical (or, at least as ethical as the trial actually done) and if the combination were superior to both monotherapy arms, would have proven that both components contribute to the claimed effects. Another alternative trial designs might also be considered is a three arm trial with: i) valsartan with ACE allowed; ii) sacubitril with ACE allowed; iii) combination with no ACE allowed. This trial may not be ethical, but if the combination was superior to the other two arms, then both components would be proven to contribute to the claimed effects. I don't know if it is ethical to have arms ii) or iii) despite the fact that arm iii) was an arm actually used in the trial reviewed here.

1.5 Evaluation of Safety

See clinical review.

1.6 Benefit-Risk Assessment (Optional)

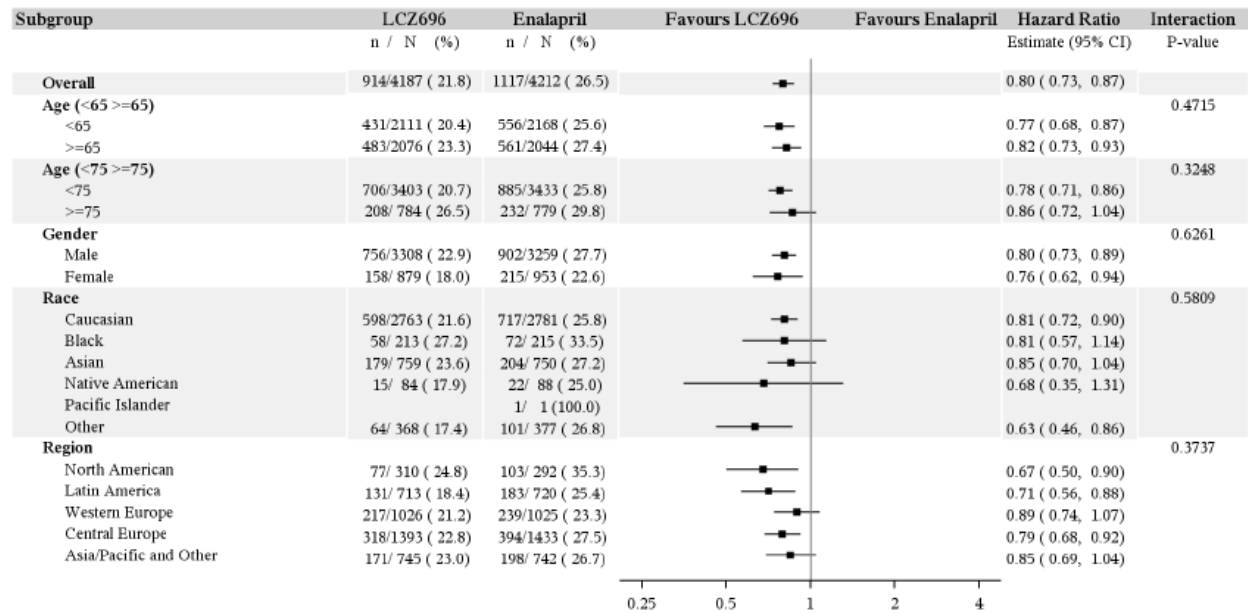
See clinical review.

FINDINGS IN SPECIAL/SUBGROUP POPULATIONS

1.7 Gender, Race, Age, and Geographic Region

The results comparing the high dose to warfarin for these subgroups for the primary endpoint are shown in Figure 6. There were no significant interactions.

Figure 6 High dose/warfarin results for subgroups defined by gender, race, age, and geographic region.



Source: Figure 11-6 of Study Report.

1.8 Other Special/Subgroup Populations

NA.

SUMMARY AND CONCLUSIONS

1.9 Statistical Issues

Because of limitations in the study design, it is not possible to know whether both components contribute to the claimed effects and this should be clearly stated in the label if this combination is approved. One-sided testing and p-values were used sometimes in the Study Report (b) (4). Two-sided p-values should be used throughout the label. P-values for the primary endpoint and the all-cause mortality endpoint were statistically significant and can be shown in the label. The components of the primary endpoint can be described in two ways. First by splitting the number of patients with the primary endpoint into mutually exclusive subsets, i.e. $377+537=914$ in the LCZ696 arm and $459+658=1117$ in the enalapril arm. Second, as shown in Table 4 by any time in the trial, whether it was the first event or not. The number of HF hospitalization events in both ways of counting will be identical, by definition. But, that is not obvious to everyone and both should be reported in the label for clarity.

1.10 Collective Evidence

NA

1.11 Conclusions and Recommendations

I don't think the combination should be approved. I don't want to prevent people from being able to purchase the components either since the combination was better than enalapril on all-cause mortality. I think there is a case for approving the sacubitril component with an indication for reducing CV mortality only. The most logical explanation for the observed mortality benefit is that sacubitril reduces mortality. It remains unknown whether the entire benefit on HF hospitalization could be explained by the valsartan component. We cannot force the company to manufacture or market sacubitril monotherapy, but that should be left to the company to decide. It is unfair to force people to buy the combination if they want to take only one component when there is no evidence that both components contribute to the claimed effect. Valsartan is approved for heart failure already and available to the public. People might reasonably choose to buy valsartan and sacubitril or they might also choose sacubitril alone or with an ACE or another ARB. Approving sacubitril as monotherapy is a less extreme option than approving the combination because approving the combination means we believe both components contribute to the mortality effect.

1.12 Labeling Recommendations (as applicable)

Because of limitations in the study design, it is not possible to know whether both components contribute to the claimed effects and this should be clearly stated in the label if this combination is approved. One-sided testing and p-values were used sometimes in the Study Report (b) (4). Two-sided p-values should be used throughout the label. P-values for the primary endpoint and the all-cause mortality endpoint were statistically significant and can be shown in the label. The components of the primary endpoint can be described in two ways. First by splitting the number of patients with the primary endpoint into mutually exclusive subsets, i.e. $377+537=914$ in the LCZ696 arm and $459+658=1117$ in the enalapril arm. Second, as shown in Table 4 by any time in the trial, whether it was the first event or not. The number of HF hospitalization events in both ways of counting will be identical, by definition. But, that is not obvious to everyone and both should be reported in the label for clarity.

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/s/

JOHN P LAWRENCE
05/18/2015

KOOROS MAHJOOB
05/20/2015



U.S. Department of Health and Human Services
Food and Drug Administration
Center for Drug Evaluation and Research
Office of Translational Sciences
Office of Biostatistics

STATISTICAL REVIEW AND EVALUATION CARCINOGENICITY STUDIES

NDA/BLA #: NDA 207620/S-0001 (IND 104628)

Drug Name: LCZ696 (sacubitril/valsartan) Tablets (AHU377)

Indication(s): Treatment of heart failure (NYHA class II-IV) (b) (4)

Applicant: Novartis Pharmaceuticals Canada Inc.
385 boul. Bouchard, Dorval, Quebec, H9S 1A9, Canada

Laboratory: (b) (4)
Novartis Pharmaceuticals Corporation (Novartis)

Date(s): Received 10/29/2014

Documents Reviewed: Study pcs-r0870373 (rats) and Study pcs-r0870374 (mice) reports and electronic datasets submitted with the electronic submission on 10/29/2014 (via S-0001).

Review Priority: Priority Review

Biometrics Division: Division of Biometrics VI

Statistical Reviewer: Feng Zhou, M.S.

Concurring Reviewers: Mohammad Atiar Rahman, Ph. D.

Team Leader: Karl Lin, Ph.D.

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Project Manager: Alexis Childers, Senior Regulatory health Project Manager

Keywords: Carcinogenicity, Dose response

Table of Contents

1	SUMMARY.....	3
2	BACKGROUND.....	3
3	RAT STUDY	4
3.1	SPONSOR'S ANALYSES	4
3.1.1	Survival Analysis	4
3.1.2	Tumor Data Analysis	5
3.2	REVIEWER'S ANALYSES	6
3.2.1	Survival Analysis	6
3.2.2	Tumor Data Analysis	7
4	MOUSE STUDY.....	8
4.1	SPONSOR'S ANALYSES	9
4.1.1	Survival Analysis	9
4.1.2	Tumor Data Analysis	9
4.2	REVIEWER'S ANALYSES	10
4.2.1	Survival Analysis	10
4.2.2	Tumor Data Analysis	10
5	CONCLUSION.....	10
6	APPENDIX	12
	Table 1A: Intercurrent Mortality – Male Rats	12
	Table 1B: Intercurrent Mortality - Female Rats	12
	Table 2A: Intercurrent Mortality - Male Mice.....	12
	Table 2B: Intercurrent Mortality – Female Mice.....	12
	Table 3A: Intercurrent Mortality Comparison – Male Rats	13
	Table 3B: Intercurrent Mortality Comparison – Female Rats.....	13
	Table 4A: Intercurrent Mortality Comparison – Male Mice.....	13
	Table 4B: Intercurrent Mortality Comparison – Female Mice.....	13
	Table 5A: Tumor Rates and P-Values for Dose Response Relationship and Pairwise Comparisons – Male Rats.....	14
	Table 5B: Tumor Rates and P-Values for Dose Response Relationship and Pairwise Comparisons – Female Rats.....	15
	Table 6A: Tumor Rates and P-Values for Dose Response Relationship and Pairwise Comparisons – Male Mice.....	16
	Table 6B: Tumor Rates and P-Values for Dose Response Relationship and Pairwise Comparisons – Female Mice.....	18
	Figure 1A: Kaplan-Meier Survival Functions for Male Rats.....	19
	Figure 1B: Kaplan-Meier Survival Functions for Female Rats.....	20
	Figure 2A: Kaplan-Meier Survival Functions for Male Mice	20
	Figure 2B: Kaplan-Meier Survival Functions for Female Mice.....	21
7	REFERENCES	21

1 Summary

This review evaluates statistically the tumorigenicity data of 2-year oral carcinogenicity studies of AHU377 in rats and mice. The review analyzes the dose-response relationship of tumor incidence and mortality (including tumor-related mortality). The review concludes that AHU377 had no effect on survival or tumor incidence in either species.

Rat Study: Rats (50/sex/dose) were dosed by oral gavage with AHU377 daily for up to 104 weeks. The AHU377 dose was 0, 50, 150, or 400-mg/kg/day in the control (C), low (LD), mid (MD) and high-dose (HD) groups in males and females, respectively. The control group (0 mg/kg/day) received the vehicle only.

Survival analysis did not show a statistically significant dose response relationship in mortality in either sex. The pairwise comparisons also did not show statistically significant increased mortality in any treated group in either sex. The respective survival rate in the C, LD, MD, and HD groups at the termination (Week 105) were 74%, 76%, 70%, and 62% in males and 60%, 60%, 68%, and 66% in females.

There was no statistically significant dose-response relationship in tumor incidence in male rats. The LD female rats showed a statistically significant increase in the incidence of uterus endometrial polyp (alone or in combination with endometrial sarcoma) when compared with the control in the pairwise comparison, but the overall analysis failed to show any dose response relationship for this tumor type.

Mouse Study: Mice (70/sex/dose) were dosed by oral gavage with AHU377 daily for up to 104 weeks. The AHU377 dose was 0, 150, 400, or 1200-mg/kg/day in the C, LD, MD and HD groups in males and females, respectively. The control group (0 mg/kg/day) received the vehicle only.

Survival analysis did not show a statistically significant dose response relationship in mortality in either sex. The pairwise comparisons also did not show statistically significant increased mortality in any treated group in either sex. The respective survival rates in the C, LD, MD and HD groups at the termination (Week 105) were 63%, 51%, 59%, and 53% in males; and 41%, 36%, 41%, and 54% in females.

The tumor analysis did not show any statistically significant dose-response relationship in tumor incidence in either sex.

2 Background

The sponsor conducted 24-month carcinogenicity studies by oral (gavage) administration in Wistar rats and CD-1 mice (one each). This review analyzed the SAS data sets of these studies received on 10/29/2014 via submission NDA207620/S0001.

The phrase "dose response relationship" refers to the linear component of the effect of treatment, and not necessarily to a strictly increasing or decreasing mortality or tumor incidence rate as dose increases. Results of this review have been discussed with the reviewing pharmacologist Dr. Timothy Link.

3 Rat Study

Study Report: pcs-r0870373—pre-clinical-study-report.pdf; **SAS data:** Tumor.xpt

This study assessed the carcinogenic potential of AHU377 in male and female Wistar rats. The test material was administered at doses of 50, 150 or 400 mg/kg/day of AHU377 in 0.5% (w/v) hydroxypropylcellulose once daily by oral gavage for at least 104 weeks. This review refers these dose groups as the low (LD), mid (MD), and high (HD) dose groups, respectively. All control animals received 0.5% Klucel HF in deionized water. The dose volume for all groups was 10 mL/kg. There were 50 rats/sex/dose. In addition, 4 rats/sex/dose served as toxicokinetic animals; an additional 10 males and 10 females were selected from the total population and used as health screen animals (group 5). The review did not discuss Group 5 because no tumor data were collected on them.

Activities performed included clinical examinations (twice daily examinations for mortality and signs of ill health or reaction to dosing, daily post dosing cage side observations, weekly detailed examinations, palpable mass examinations on main study animals), body weights, food consumption, laboratory investigations (hematology) for health screen prior to dosing initiation and for main study animals at study termination (blood smears). Post-mortem evaluations for main study animals included macroscopic and microscopic observations.

3.1 Sponsor's Analyses

3.1.1 Survival Analysis

The sponsor performed survival analysis using the Kaplan-Meier product-limit method. An overall trend test comparing all groups was conducted using a log-rank test (Mantel 1966, Peto 1974). Any animal with accidental injury that causes its death or its unscheduled sacrifice was censored in the estimation. The log-rank test was applied to the four groups in order to assess the significance of the overall group effect on mortality data. If this test was found to be significant ($p \leq 0.05$), then the significance of a dose-related trend in mortality across all groups was evaluated using Tarone's method (Tarone, 1975). Using the Multtest procedure (SAS Institute, 1999), Tarone's test was implemented as a Peto two-sided test, with all uncensored deaths coded as 2 and all censored deaths coded as 0. The corresponding arithmetic dose level scores (0, 50, 150 and 400) were used to perform this overall trend test. Furthermore, the control group was compared to each of the other three groups using a Peto two-sided test. The sign of the statistic was used to indicate if either an increase or a decrease of the mortality rate across dose levels was observed for the trend test and each pairwise comparison. When performing a pairwise comparison, only the data corresponding to the two compared groups was submitted to the statistical analysis. Each trend test and pairwise group comparison of interest was conducted at the 5% significance level. Significant trend and pairwise comparison results were reported as $p \leq 0.001$, $p \leq 0.01$ or $p \leq 0.05$, where p represents the obtained probability." The animal was classified as uncensored when death occurred within the experimental period, inclusively day 1 to day 728. Otherwise, that is, when the animal's death was accidental, as it is the case for the animals #s. 1506 and 2539, and/or occurred during the terminal sacrifice period (day 729 and over), the animal was classified as censored.

Sponsor's findings: There was no AHU377-related effect on survival. At the end of the dosing period, the survival rates ranged from 62 to 76% for males and 60 to 68% for females. The total unscheduled deaths are listed in the following table. There appeared a dose-related trend in the mortality rate in males, but none of the increase reached a statistically significant level ($p > 0.05$). Also, the mortality rate was comparable between all groups for both sexes. For each pre-terminal

decendent rat, the most probable cause of death was determined. The cause of death could not be determined for a small number of animals per group.

The Sponsor's Report of Total Unscheduled Deaths, n=50/sex/dose group)

Group no. identification	Dose level (mg/kg/day)	Survival %			
		Males	%	Females	%
1/ Control	0	13/50	74%	20/50	60%
2/ AHU377	50	12/50	76%	20/50	60%
3/ AHU377	150	15/50	70%	16/50	68%
4/ AHU377	400	19/50	62%	17/50	66%

These numbers included animals that were found dead after week 104.

[Source: page 36 of study report of pcs-r0870373—pre-clinical-study-report.pdf]

3.1.2 Tumor Data Analysis

The sponsor analyzed the tumor incidence data using the methods outlined in the paper of Peto et al. (1980) for positive dose response relationships and the Fisher exact test for pairwise comparisons of the treated groups with the control. Neoplastic findings classified as fatal and incidental were processed using the death rate method and the prevalence method, respectively. The processing of incidental tumors was done by creating a single separate interval for the time period following the experimental period (terminal sacrifice period) and by dividing the experimental period into the following fixed intervals (FDA's draft guidance for industry, 2001): weeks 1 - 52, weeks 53 - 78, weeks 79 - 92, weeks 93-104, and terminal sacrifice (TS). Using the derived outcomes from the processing of both fatal and incidental tumors, a test statistic was built to perform a global survival-adjusted trend test on tumor data observed in a "mortality dependent" context (Peto et al, 1980). For each dataset within each sex, the significance of an overall linear dose-related increase in tumor occurrence rates across the vehicle control group (group C) and the three test item treated groups LD, MD and HD were evaluated using Peto's survival-adjusted one-tailed trend test. The corresponding arithmetic dose level scores (0, 50, 150 and 400) were used to perform this overall trend test. When the overall trend test was found significant, the sequential trend test was performed at the 5% significance level. The sequential trend test was implemented by excluding successively the highest dosed group from the analysis until the one-sided Peto trend test was no longer significant. Furthermore, Peto's one tailed trend test was also used to test when the tumor rate in each of the three test item treated groups LD, MD, and HD was significantly higher than the tumor rate in the vehicle control group. When performing a pairwise comparison, only the data corresponding to the two compared groups was submitted to the statistical analysis. Following the suggestion of Lin (Lin, 1997) the discrete permutation distribution was used to compute the corresponding p-value for each statistical test performed on a data set containing 10 or less tumor occurrences. All tests were performed at the 5% significance level. However, each trend test (the overall trend and the pairwise comparison tests) were discussed in the statistical report according to the tumor prevalence classification (common or rare) and the recommendations of Lin and Rahman (Lin and Rahman, 1998).

Adjustment for multiple testing: In order to control the false positive error, the sponsor tested the common and the rare tumors at 0.005 and 0.025 significance levels, respectively (Lin, 2000) for positive dose response relationship, and 0.01 and 0.05 for pairwise comparisons. Tumors are considered as common with a background rate of $\geq 1\%$ and as rare with a background incidence of $< 1\%$.

Sponsor's findings: A statistically significant ($p = 0.0046$) positive dose response in the incidence of pituitary adenoma of the pars distalis was recorded in males. However, the pairwise comparison high dose group with control did not result in statistical significance for a common tumor ($p = 0.0153$).

The sponsor indicated that although the overall incidence of this tumor type was slightly above the historical range of this laboratory (20% to 37% for male rats), this change was considered equivocal and not likely to be an indication of carcinogenic potential of AHU377.

A few other tumors or tumor combinations reached statistical significance including follicular adenoma of the thyroid with a p-value of 0.0248 (overall trend test) in females only, combined hemangioma/hemangiosarcoma with a p-value of 0.0085 in males only (pairwise comparison test: group C vs group MD) and uterus endometrial stroma polyp in females (pairwise comparison test was significant (group C vs group LD) with a p-value of 0.0077) were observed. Follicular adenoma in the thyroid is a common tumor, the p-value did not reach statistical significance for common tumors and this increased incidence in females was therefore considered of no toxicological significance. Since the overall trend test was not significant and the pairwise comparison test with the high dose group did not reach statistical significance for the hemangioma/hemangiosarcoma combination in males and for endometrial polyp in the uterus of females, and the incidence of the uterine endometrial polyp was within range of the historical control data at this laboratory (incidence between 6% to 22% for uterine endometrial polyp. The sponsor mentioned that no historical control data are available for combined hemangioma/hemangiosarcoma), the higher incidence of these tumor/tumor combinations in the intermediate or low dose group were interpreted to be incidental and of no toxicological significance. All other tumor incidences recorded were within the range of spontaneous occurrence reported for aged Wistar Hannover rats, were generally randomly distributed in control and treated groups and/or lacked a dose-related pattern. In addition, no increase in the incidence of any hyperplastic lesions was noted

The sponsor concluded that AHU377 given orally to Wistar Han rats at 50, 150 and 400 mg/kg/day for a minimum of 104 consecutive weeks was well tolerated. There were no AHU377-related effects on survival rate, clinical signs, body weights and food consumption. No evidence of any carcinogenic effect was seen in male or female rats given up to 400 mg/kg/day of AHU377. No gross or microscopic changes were seen in any organs or tissues examined that were attributed to the oral gavage administration of AHU377 up to 400 mg/kg/day.

3.2 Reviewer's Analyses

To verify the sponsor's analyses and to perform additional analyses suggested by the reviewing pharmacologist, this reviewer performed survival and tumor data analyses using data submitted electronically in NDA 207-620 on 10/29/2014.

The pharmacologist requested statistical evaluations of the following:

- Is there any difference in the survival (or death) among groups; if yes, whether the difference is statistically significant.
- Is there any difference in the incidences of any neoplastic finding among groups, if yes, whether the difference is statistically significant.

3.2.1 *Survival Analysis*

The survival distributions of rats in all treatment groups were estimated using the Kaplan-Meier product limit method. For control, low, medium, and high dose groups, the dose response relationship was tested using the likelihood ratio test and the homogeneity of survival distributions was tested using the log-rank test. The Kaplan-Meier curves for survival rates are given in Figures 1A and 1B in the appendix for male and female rats, respectively. The intercurrent mortality data are given in Tables 1A and 1B in the appendix for male and female rats, respectively. Results of the tests for dose response relationship and

homogeneity of survivals, are given in Tables 3A and 3B in the appendix for male and female rats, respectively.

Reviewer's findings: This reviewer's analysis showed the numbers (percent) of death were 13 (26%), 12 (24%), 15 (30%), and 19 (38%) in male rats and 20 (40%), 20 (40%), 16 (32%), and 17 (34%) in female rats in the C, LD, MD, and HD groups, respectively. The tests didn't show a statistically significant dose response relationship in mortality across control and treated groups in male or female rats. The pairwise comparisons did not show statistically significant increased mortality in the treated groups compared to the control in either sex.

3.2.2 Tumor Data Analysis

The tumor data were analyzed for dose response relationships and pairwise comparisons of control group with each of the treated groups. Both the dose response relationship tests and pairwise comparisons were performed using the Poly-k method described in the paper of Bailer and Portier (1988) and Bieler and Williams (1993). In this method an animal that lives the full study period ($w_{\max}$) or

dies before the terminal sacrifice but develops the tumor type being tested gets a score of $s_h = 1$. An animal that dies at week w_h without developing the tumor before the end of the study gets a score of

$s_h = \left(\frac{w_h}{w_{\max}} \right)^k < 1$. The adjusted group size is defined as $\sum s_h$. As an interpretation, an animal with score

$s_h = 1$ can be considered as a whole animal while an animal with score $s_h < 1$ can be considered as a partial animal. The adjusted group size $\sum s_h$ is equal to N (the original group size) if all animals live up to the end of the study or if each animal that dies before the terminal sacrifice develops at least one tumor, otherwise the adjusted group size is less than N. These adjusted group sizes are then used for the dose response relationship (or the pairwise) tests using the Cochran-Armitage test. One critical point for Poly-k test is the choice of the appropriate value of k, which depends on the tumor incidence pattern with the increased dose. For long term 104 week standard rat and mouse studies, a value of k=3 is suggested in the literature. Hence, this reviewer used k=3 for the analysis of this data. For the calculation of p-values the exact permutation method was used. The tumor rates and the p-values of the tested tumor types are listed in Tables 5A and 5B in the appendix for male and female rats, respectively.

Multiple testing adjustment: For the adjustment of multiple testing of dose response relationship, the FDA guidance for the carcinogenicity study design and data analysis suggests the use of test levels $\alpha=0.005$ for common tumors and $\alpha=0.025$ for rare tumors for a submission with two species, and a significance level $\alpha=0.01$ for common tumors and $\alpha=0.05$ for rare tumors for a submission with one species in order to keep the false-positive rate at the nominal level of approximately 10%. A rare tumor is defined as one in which the published spontaneous tumor rate is less than 1%. For multiple pairwise comparisons of treated group with control the FDA guidance the suggested the use of test levels $\alpha=0.01$ for common tumors and $\alpha=0.05$ for rare tumors, in order to keep the false-positive rate at the nominal level of approximately 10% for both submissions with two or one species.

It should be noted that the FDA guidance for multiple testing for dose response relationship is based on a publication by Lin and Rahman (1998). In this work the authors investigated the use of this rule for Peto analysis. However, in a later work Lin and Rahman (2008) showed that this rule for multiple testing for dose response relationship is also suitable for Poly-K tests.

Reviewer's findings: Following two tables display the tumor types showed p-values less than or equal to 0.05 either for dose response relationship or pairwise comparisons of treated groups and control.

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Tumor Types with P-Values ≤ 0.05 for Dose Response Relationship or Pairwise Comparisons of Treated Groups and Combined Controls in Male Rats

Organ Name	Tumor Name	0	50	150	400	P-Value			
		mg/kg/day C N=50	mg/kg/day LD N=50	mg/kg/day MD N=50	mg/kg/day HD N=50	Dose Response	C vs. LD	C vs. MD	C vs. HD
PITUITARY	Adenoma: pars distal	13	12	15	21	0.0136	0.4754	0.3594	0.0484
Combined organ	C Hemangioma+Hemangi osarcoma	3	6	10	4	0.4468	0.2315	0.0255	0.4351

*Indicted the significant at 0.005 or 0.025 alpha levels.

Tumor Types with P-Values ≤ 0.05 for Dose Response Relationship or Pairwise Comparisons of Treated Groups and Combined Controls in Female Rats

Organ Name	Tumor Name	0	50	150	400	P-Value			
		mg/kg/day C N=50	mg/kg/day LD N=50	mg/kg/day MD N=50	mg/kg/day HD N=50	Dose Response	C vs. LD	C vs. MD	C vs. HD
MAMMARY GLAND	Fibroadenoma	13	9	15	19	0.0378	0.7304	0.3821	0.1537
THYROID	Adenoma: follicular	1	0	1	4	0.0250	0.4941	0.7471	0.1947
UTERUS	Sarcoma: endometrial	0	2	0	0	0.7602	0.2470	.	.
	Polyp: endometrial s	1	9	6	5	0.4245	0.0074*	0.0548	0.1120
	C Endometrial	1	11	6	5	0.5153	0.0017*	0.0548	0.1120
Combined organ	C Hemangioma+Hemangi osarcoma	0	1	1	3	0.0495	0.5000	0.5059	0.1339

*Indicted the significant at 0.005 or 0.025 alpha levels.

Based on the criteria of adjustment for multiple testing discussed above, none of the observed tumors was considered to have a statistically significant positive dose response relationship in either sex. In female rats the pairwise comparison showed statistically significant increased incidence of uterus endometrial polyp in low dose group compared to the control ($p=0.0074<0.01$) and the combination of Sarcoma: endometrial and Polyp: endometrial in Uterus in low dose group compared to the control ($p=0.0017<0.01$). However, the dose response relationship for this tumor type was not statistically significant.

Reviewer's comment: Note that the sponsor's p-value for dose response relationship test for the incidence of pituitary adenoma of the pars distalis in male rats is 0.0046, while that calculated by this reviewer is 0.0136. This discrepancy between the p-values is due to the difference methodologies used by the sponsor (Peto method) and this reviewer (Poly-K).

4 Mouse Study

Study Report: PCS-R0870374.pdf; SAS data: Tumor.xpt

This study assessed the carcinogenic potential of AHU377 in male and female CD-1 mice. The test material was administered at doses of 150, 400, or 1200 mg/kg/day of AHU377 in 0.5% methylcellulose once daily by oral gavage for approximately 2 years. This review refers these dose groups as the low (LD), mid (MD), and high (HD) dose groups, respectively. All control animals received 0.5% methylcellulose in deionized water. The dose volume was 10 mL/kg. There were 70 mice /sex/dose. The toxicokinetic arm of the study contained 4 groups (one control group and three AHU377-treated groups) of 8 females and 8 males each.

Activities performed included clinical examinations (twice daily examinations for mortality and signs of ill health or reaction to dosing, daily post dosing cage side observations, weekly detailed examinations, palpable mass examinations on main study animals), body weights, food consumption,

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laboratory investigations (hematology) for health screen animals prior to dosing initiation and for main animals at study termination (blood smears). Post-mortem evaluations for main study animals included macroscopic observations and microscopic observations.

4.1 Sponsor's Analyses

4.1.1 Survival Analysis

The sponsor performed survival analysis using the same methodologies that were used in the rat study. Any animal with accidental injury that caused its death or its unscheduled sacrifice was censored in the estimation. The male high dose group was excluded from the sponsor's overall analysis due to the dosing of the animals in this group stopped after day 319. The animal was classified as uncensored when death occurred within the experimental period, inclusively day 1 to day 728. Otherwise, that is, when the animal's death was accidental, as it is the case for the animals #s. 1011, 2016, 2532, 3534, 3545, 3558, 4010, 4011, 4025, 4070, and 4515, and/or occurred during the terminal sacrifice period (day 729 and over), the animal was classified as censored.

Sponsor's findings: The sponsor mentioned that the mortality rate was comparable among all groups for both sexes. There was no AHU377-related effect on survival. At the end of the dosing period, the survival rates ranged from 51 to 63% for males and 36 to 54% for females.

The Sponsor's Report of Total Unscheduled Deaths, n=70/sex/group)

Study week	Males				Females			
	Dose (mg/kg/day)				Dose (mg/kg/day)			
	0	150	400	1200	0	150	400	1200
16	70 (100%)	70 (100%)	70 (100%)	70 (100%)	70 (100%)	70 (100%)	70 (100%)	70 (100%)
24	70 (100%)	70 (100%)	69 (99%)	70 (100%)	69 (99%)	69 (99%)	69 (99%)	70 (100%)
52	70 (100%)	66 (94%)	65 (93%)	66 (94%)	67 (96%)	63 (90%)	66 (94%)	66 (94%)
80	63 (90%)	54 (77%)	53 (76%)	56 (80%)	58 (83%)	48 (69%)	56 (80%)	54 (77%)
Term	44 (63%)	36 (51%)	41 (59%)	37 (53%)	29 (41%)	25 (36%)	29 (41%)	38 (54%)

[Source: page 38 of study report of pcs-r0870374-pre-clinical-study-report.pdf]

4.1.2 Tumor Data Analysis

The sponsor analyzed the tumor incidence data using the same methodologies that were used in the rat study. Notice that, due to the dosing of the animals in male high dose group was stopped after day 319, the sponsor excluded male high dose group from the tumor analysis.

For each dataset within each sex, the significance of an overall linear dose-related increase in tumor occurrence rates, across groups 1, 2, 3 and 4 for the females and across groups 1, 2 and 3 for the males, was evaluated using Peto's survival-adjusted one-tailed trend test. The corresponding arithmetic dose level scores that were used to perform this overall trend test were given by 0, 150, 400 and 1200.

Sponsor's findings:

Clinically observed palpable masses, which were present during the dosing phase and at necropsy, were generally noted in a similar proportion of animals from control and AHU377-treated groups. There was no evidence of statistically significant carcinogenic effect of AHU377 in male or female

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mice at histopathology examination. The sponsor mentioned that the tumor incidences recorded were within the range of spontaneous occurrence reported for aged CD-1 mice.

The sponsor concluded that AHU377 given orally to CD1 mice at 150, 400 and 1200 mg/kg/day for a minimum of 104 consecutive weeks was well tolerated. There were no AHU377-related effects on survival rate, clinical signs. No evidence of any carcinogenic effect was seen in male or female mice given up to 1200 mg/kg/day of AHU377.

Please notice that, the sponsor did not use the data from male high dose group in their analysis.

4.2 Reviewer's Analyses

To verify the sponsor's analyses and to perform additional analyses suggested by the reviewing pharmacologist, this reviewer performed survival and tumor data analyses using data submitted electronically in submission NDA 207-620 on 10/29/2014.

4.2.1 Survival Analysis

The Kaplan-Meier curves for survival rates of all treatment groups are given in Figures 2A and 2B in the appendix for male and female mice, respectively. The intercurrent mortality data of all treatment groups are given in Tables 2A and 2B in the appendix for male and female mice, respectively. Results of the tests for dose response relationship and homogeneity of survivals for control, low, medium, and high dose groups are given in Tables 4A and 4B in the appendix for male and female mice, respectively.

Reviewer's findings: This reviewer's analysis showed 26(37%), 34 (49%), 29 (41%), and 33 (47%) number (percent) of deaths in male mice, and 41 (59%), 45 (64%), 41 (59%), and 32 (46%) number (percent) of deaths in female mice the C, LD, MD, and HD groups, respectively. The tests did not show any statistically significant dose response relationship in mortality across control and treated groups in either sex mice. The pairwise comparisons did not show statistically significant increased mortality in the treated groups compared to the control in either male or female mice.

4.2.2 Tumor Data Analysis

The tumor data were analyzed for dose response relationships and pairwise comparisons of control group with each of the treated groups using the same method that was used for the rats study. The tumor rates and the p-values of the tested tumor types are listed in Tables 6A and 6B in the appendix for male and female mice, respectively.

Reviewer's findings: Based on the criteria of adjustment for multiple testing discussed in the rat data analysis section, none of the observed tumors was considered to have a statistically significant positive dose response relationship or increased incidence in any treated group compared to their control in either sex of mice.

5 Conclusion

This review evaluates statistically the tumorigenicity data of 2-year oral carcinogenicity studies of AHU377 in rats and mice. The review analyzes the dose-response relationship of tumor incidence and mortality (including tumor-related mortality). The review concludes that AHU377 had no effect on survival or tumor incidence in either species.

Rat Study: Rats (50/sex/dose) were dosed by oral gavage with AHU377 daily for up to 104 weeks. The AHU377 dose was 0, 50, 150, or 400-mg/kg/day in the C, LD, MD and HD groups in males and females, respectively. The control group (0 mg/kg/day) received the vehicle only.

Survival analysis did not show a statistically significant dose response relationship in mortality in either sex. The pairwise comparisons also did not show statistically significant increased mortality in any treated group in either sex. The respective survival rate in the C, LD, MD, and HD groups at the termination (Week 105) were 74%, 76%, 70%, and 62% in males and 60%, 60%, 68%, and 66% in females.

There was no statistically significant dose-response relationship in tumor incidence in male rats. The LD female rats showed a statistically significant increase in the incidence of uterus endometrial polyp (alone or in combination with endometrial sarcoma) when compared with the control in the pairwise comparison, but the overall analysis failed to show any dose response relationship for this tumor type.

Tumor Incidence Rates in Female Rats

Organ Name	Tumor Name	0 mg/kg/day C N=50	50 mg/kg/day LD N=50	150 mg/kg/day MD N=50	400 mg/kg/day HD N=50	Dose Response	P-Value		
							C vs. LD	C vs. MD	C vs. HD
UTERUS	Sarcoma: endometrial	0	2	0	0	0.7602	0.2470	.	.
	Polyp: endometrial s	1	9	6	5	0.4245	0.0074*	0.0548	0.1120
	C Endometrial	1	11	6	5	0.5153	0.0017*	0.0548	0.1120

*Indicted the significant at 0.005 or 0.025 alpha levels.

Mouse Study: Mice (70/sex/dose) were treated with 0, 150, 400, or 1200-mg/kg/day AHU377 for up to 104 weeks. The control group (0 mg/kg/day) received the vehicle only.

Survival analysis didn't show a statistically significant dose response relationship in mortality in male or female mice. The pairwise comparisons also did not show statistically significant increased mortality in any treated group in either sex. The respective survival rates in the C, LD, MD and HD groups at the termination (Week 105) were 63%, 51%, 59%, and 53% in males; and 41%, 36%, 41%, and 54% in females.

The tumor analysis did not show any statistically significant dose-response relationship in tumor incidence in either sex.

Feng Zhou, M.S.
Mathematical Statistician

Secondary Reviewer: Atiar Mohammad Rahman, Ph.D.

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6 Appendix

Table 1A: Intercurrent Mortality – Male Rats

Week	0 mg/kg/day (n=50)		50 mg/kg/day (n=50)		150 mg/kg/day (n=50)		400 mg/kg/day (n=50)	
	No. of Death	Cum. %	No. of Death	Cum. %	No. of Death	Cum. %	No. of Death	Cum. %
0 - 52	.	.	.	.	1	2.00	1	2.00
53 - 78	2	4.00	2	4.00	4	10.00	5	12.00
79 - 91	3	10.00	7	18.00	7	24.00	7	26.00
92 - 104	8	26.00	3	24.00	3	30.00	6	38.00
Ter. Sac.	37	74.00	38	76.00	35	70.00	31	62.00

* Cum. %: Cumulative percentage except for Ter. Sac.

Table 1B: Intercurrent Mortality - Female Rats

Week	0 mg/kg/day (n=50)		50 mg/kg/day (n=50)		150 mg/kg/day (n=50)		400 mg/kg/day (n=50)	
	No. of Death	Cum. %	No. of Death	Cum. %	No. of Death	Cum. %	No. of Death	Cum. %
0 - 52	.	.	1	2.00	1	2.00	.	.
53 - 78	3	6.00	4	10.00	5	12.00	2	4.00
79 - 91	8	22.00	6	22.00	3	18.00	5	14.00
92 - 104	9	40.00	9	40.00	7	32.00	10	34.00
Ter. Sac.	30	60.00	30	60.00	34	68.00	33	66.00

* Cum. %: Cumulative percentage except for Ter. Sac.

Table 2A: Intercurrent Mortality - Male Mice

Week	0 mg/kg/day (n=70)		150 mg/kg/day (n=70)		400 mg/kg/day (n=70)		1200 mg/kg/day (n=70)	
	No. of Death	Cum. %	No. of Death	Cum. %	No. of Death	Cum. %	No. of Death	Cum. %
0 - 52	.	.	4	5.71	5	7.14	4	5.71
53 - 78	5	7.14	12	22.86	11	22.86	9	18.57
79 - 91	10	21.43	7	32.86	5	30.00	6	27.14
92 - 104	11	37.14	11	48.57	8	41.43	14	47.14
Ter. Sac.	44	62.86	36	51.43	41	58.57	37	52.86

* Cum. %: Cumulative percentage except for Ter. Sac.

Table 2B: Intercurrent Mortality – Female Mice

Week	0 mg/kg/day (n=70)		150 mg/kg/day (n=70)		400 mg/kg/day (n=70)		1200 mg/kg/day (n=70)	
	No. of Death	Cum. %	No. of Death	Cum. %	No. of Death	Cum. %	No. of Death	Cum. %
0 - 52	3	4.29	7	10.00	4	5.71	4	5.71
53 - 78	7	14.29	15	31.43	10	20.00	10	20.00
79 - 91	16	37.14	10	45.71	7	30.00	7	30.00
92 - 104	15	58.57	13	64.29	20	58.57	11	45.71
Ter. Sac.	29	41.43	25	35.71	29	41.43	38	54.29

* Cum. %: Cumulative percentage except for Ter. Sac.

Table 3A: Intercurrent Mortality Comparison – Male Rats

Test	Statistic	P-Value
Dose-Response	Likelihood Ratio	0.0793
Homogeneity	Log-Rank	0.3339

Table 3B: Intercurrent Mortality Comparison – Female Rats

Test	Statistic	P-Value
Dose-Response	Likelihood Ratio	0.3734
Homogeneity	Log-Rank	0.7659

Table 4A: Intercurrent Mortality Comparison – Male Mice

Test	Statistic	P-Value
Dose-Response	Likelihood Ratio	0.5747
Homogeneity	Log-Rank	0.4321

Table 4B: Intercurrent Mortality Comparison – Female Mice

Test	Statistic	P-Value
Dose-Response	Likelihood Ratio	0.0710
Homogeneity	Log-Rank	0.1387

Table 5A: Tumor Rates and P-Values for Dose Response Relationship and Pairwise Comparisons – Male Rats

Organ Name	Tumor Name	0	50	150	400	P-Value			
		mg/kg/day C N=50	mg/kg/day LD N=50	mg/kg/day MD N=50	mg/kg/day HD N=50	Dose Response	C vs. LD	C vs. MD	C vs. HD
ADRENAL	Adenoma: cortical	4	3	2	0	0.9763	0.4876	0.6401	0.9267
	Benign pheochromocyt	0	2	1	3	0.0675	0.2418	0.4831	0.1006
	Carcinoma: cortical	0	0	2	0	0.4740	.	0.2306	.
	Ganglioneuroma	0	0	0	1	0.2343	.	.	0.4713
	C CORTICAL ADEN+CAR	4	3	4	0	0.9582	0.4876	0.6177	0.9267
BONE MISC.	Osteosarcoma	0	0	0	1	0.2343	.	.	0.4713
BRAIN	Benign granular cell	2	0	0	0	0.9320	0.7473	0.7357	0.7233
	Malignant astrocytom	0	1	0	0	0.4800	0.4945	.	.
	Malignant granular c	1	0	0	0	0.7371	0.4945	0.4831	0.4713
	Malignant mixed glio	0	0	0	1	0.2343	.	.	0.4713
EP D DYMIS	Mesothelioma(B)	1	0	0	0	0.7371	0.4945	0.4831	0.4713
FAT	Lipoma	2	0	0	1	0.5191	0.7473	0.7357	0.4564
HEART	Malignant schwannoma	0	0	0	1	0.2386	.	.	0.4773
JEJUNUM	Benign schwannoma	0	0	0	1	0.2343	.	.	0.4713
	Leiomyosarcoma	1	0	0	0	0.7371	0.4945	0.4831	0.4713
K DNEY	Lipoma	0	0	1	0	0.4800	.	0.4831	.
	Papilloma: transitio	0	0	1	0	0.4800	.	0.4831	.
LIVER	Adenoma: hepatocellu	3	2	4	6	0.0563	0.4895	0.4619	0.1877
LUNG	Adenoma: alveolar/br	2	2	0	1	0.7009	0.6834	0.7357	0.4564
	Carcinoma: alveolar/	1	0	0	0	0.7330	0.4891	0.4778	0.4659
	C ALVEOLAR ADEN+CAR	3	2	0	1	0.8099	0.4792	0.8620	0.6384
LYMPH NODE	Hemangioma	0	0	1	1	0.1697	.	0.4831	0.4713
MESENTERIC LN	Hemangioma	1	2	2	2	0.3188	0.4917	0.4744	0.4564
	Hemangiosarcoma	1	2	4	1	0.5024	0.4917	0.1600	0.7233
MUSCLE MISC.	Hemangiosarcoma	0	0	1	0	0.4800	.	0.4831	.
PANCREAS	Adenoma: acinar cell	0	0	0	2	0.0539	.	.	0.2192
	Adenoma: islet cell	0	1	2	1	0.2916	0.4945	0.2306	0.4713
	Carcinoma: islet cel	1	1	1	0	0.7096	0.7473	0.7357	0.4713
	C ISLET ADEN+CAR	1	2	3	1	0.5120	0.4917	0.2830	0.7233
PARATHYROID GL	Adenoma	0	1	1	0	0.4740	0.4945	0.4831	.
PITUITARY	Adenoma: pars distal	13	12	15	21	0.0136	0.4754	0.3594	0.0484
	Adenoma: pars interm	1	0	2	1	0.3367	0.4945	0.4744	0.7233
SALIV.GL. MANDI	Adenoma	0	1	0	0	0.4800	0.4945	.	.
SK N MISC.	Adenoma: basal cell	1	2	0	0	0.8778	0.4917	0.4831	0.4713
	Adenoma: sebaceous	0	2	0	0	0.7310	0.2418	.	.
	Carcinoma: basal cel	0	1	0	0	0.4800	0.4945	.	.
	Carcinoma: squamous	2	0	1	1	0.4780	0.7418	0.4663	0.4483
	Fibroma	0	1	0	1	0.2908	0.4945	.	0.4713
	Hemangioma	0	0	1	0	0.4800	.	0.4831	.
	Keratoacanthoma	10	2	8	5	0.6546	0.9852	0.5405	0.8265
	Papilloma: squamous	0	1	1	2	0.1022	0.4945	0.4831	0.2192
	C BASAL ADEN+CAR	1	3	0	0	0.9159	0.3000	0.4831	0.4713
	C SQUAMOUS ADEN+CAR+	12	3	9	8	0.4604	0.9863	0.6039	0.6823
	SP NAL CORD CER	0	0	0	1	0.2343	.	.	0.4713
STOMACH	Carcinoma: squamous	0	0	0	1	0.2343	.	.	0.4713
	Papilloma: squamous	0	0	0	1	0.2343	.	.	0.4713
	C SQUAMOUS ADEN+CAR	0	0	0	2	0.0539	.	.	0.2192

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Organ Name	Tumor Name	0	50	150	400	P-Value			
		mg/kg/day C N=50	mg/kg/day LD N=50	mg/kg/day MD N=50	mg/kg/day HD N=50	Dose Response	C vs. LD	C vs. MD	C vs. HD
SUBCU. TISSUE	Fibroma	4	5	6	3	0.6212	0.4858	0.3268	0.4351
	Fibrosarcoma	1	0	0	0	0.7371	0.4945	0.4831	0.4713
	Hemangiosarcoma	1	2	1	0	0.8070	0.4917	0.7357	0.4713
	Lipoma	2	2	2	1	0.6441	0.6750	0.6573	0.4483
	Malignant schwannoma	2	1	0	0	0.9274	0.4835	0.7301	0.7176
SYSTEMIC	Malignant lymphoma	1	2	2	2	0.3154	0.5000	0.4831	0.4564
TESTIS	Adenoma: interstitia	1	3	5	0	0.7844	0.3000	0.0867	0.4713
THORAX	Mesothelioma(M)	0	1	0	0	0.4800	0.4945	.	.
THYMUS	Benign thymoma	2	3	1	1	0.7440	0.4895	0.4744	0.4564
	Malignant thymoma	0	1	0	0	0.4800	0.4945	.	.
THYROID	Adenoma: C-cell	7	9	8	7	0.4907	0.3554	0.4244	0.5034
	Adenoma: follicular	5	0	1	3	0.4525	0.9705	0.8824	0.5765
	Carcinoma: C-cell	1	1	1	1	0.4829	0.7473	0.7357	0.7233
	Carcinoma: follicula	0	1	0	1	0.2908	0.4945	.	0.4713
	C C CELL ADEN+CAR	8	10	9	8	0.4771	0.3573	0.4186	0.4883
	C FOLLICULAR ADEN+CA	5	1	1	4	0.3337	0.8934	0.8824	0.4261
TONGUE	Hemangiosarcoma	0	0	1	0	0.4800	.	0.4831	.
URINARY BLADDER	Carcinoma: transitio	0	0	0	1	0.2343	.	.	0.4713
Combined Organ	C Hemangioma+ Hemangiosarcoma	3	6	10	4	0.4468	0.2315	0.0255	0.4351

Table 5B: Tumor Rates and P-Values for Dose Response Relationship and Pairwise Comparisons – Female Rats

Organ Name	Tumor Name	0	50	150	400	P-Value			
		mg/kg/day C N=50	mg/kg/day LD N=50	mg/kg/day MD N=50	mg/kg/day HD N=50	Dos Response	C vs. LD	C vs. MD	C vs. HD
ADRENAL	Adenoma: cortical	4	2	2	3	0.5368	0.6507	0.6507	0.5257
	Benign pheochromocyt	0	0	1	0	0.2632	.	0.5000	.
	Carcinoma: cortical	1	0	1	0	0.6388	0.5000	0.7530	0.5172
	Ganglioneuroma	0	0	0	1	0.2632	.	.	0.5172
	C CORTICAL ADEN+CAR	5	2	3	3	0.6459	0.7737	0.6302	0.6688
BONE MISC.	Osteosarcoma	0	1	0	0	0.5088	0.5000	.	.
BRA N	Benign granular cell	1	0	0	0	0.7544	0.5000	0.5000	0.5172
	Benign pineal gland	0	0	1	0	0.2632	.	0.5000	.
	Malignant meningioma	0	1	0	0	0.5058	0.5059	.	.
	Malignant oligodendr	1	0	0	0	0.7500	0.4941	0.4941	0.5114
FAT	Lipoma	1	0	1	0	0.6344	0.4941	0.7471	0.5114
	Liposarcoma	0	1	0	0	0.5088	0.5000	.	.
K DNEY	Carcinoma: tubular c	0	1	0	1	0.3282	0.5000	.	0.5172
	Lipoma	0	0	0	1	0.2632	.	.	0.5172
LIVER	Adenoma: hepatocellu	3	5	3	5	0.3588	0.3563	0.6621	0.3964
	Carcinoma: hepatocel	0	0	0	1	0.2632	.	.	0.5172
	Hemangiosarcoma	0	0	0	1	0.2632	.	.	0.5172
	C HEPATOCELLULAR ADE	3	5	3	6	0.2426	0.3563	0.6621	0.2900
LYMPH NODE	Carcinoma: metastasi	1	0	0	0	0.7500	0.4941	0.4941	0.5114
MAMMARY GLAND	Adenocarcinoma	6	2	6	2	0.8563	0.8518	0.6048	0.8807
	Adenoma	1	0	1	0	0.6388	0.5000	0.7530	0.5172
	Fibroadenoma	13	9	15	19	0.0378	0.7304	0.3821	0.1537

Organ Name	Tumor Name	0	50	150	400	P-Value			
		mg/kg/day C N=50	mg/kg/day LD N=50	mg/kg/day MD N=50	mg/kg/day HD N=50	Dos Response	C vs. LD	C vs. MD	C vs. HD
	C ADEN+CAR+FIBROADEN	17	10	20	20	0.1267	0.8937	0.3018	0.3999
MESENTERIC LN	Hemangioma	0	0	0	1	0.2632	.	.	0.5172
	Hemangiosarcoma	0	1	0	0	0.5088	0.5000	.	.
OVARY	Adenoma: sertoliform	1	0	0	0	0.7544	0.5000	0.5000	0.5172
	Adenoma: tubulostrom	1	2	2	1	0.5810	0.5000	0.5000	0.2646
	Carcinoma: yolk sac	1	0	0	0	0.7500	0.4941	0.4941	0.5114
PANCREAS	Adenoma: islet cell	0	1	0	1	0.3282	0.5000	.	0.5172
PARATHYRO D GL	Adenoma	0	0	0	1	0.2632	.	.	0.5172
PITUITARY	Adenoma: pars distal	33	32	32	30	0.7750	0.5550	0.5000	0.6661
	Adenoma: pars interm	0	0	1	1	0.2018	.	0.5000	0.5227
	Carcinoma: pars dist	1	0	0	0	0.7500	0.4941	0.4941	0.5114
	Carcinoma: pars inte	1	0	0	0	0.7544	0.5000	0.5000	0.5172
	C DISTALIS ADEN+CAR	34	32	32	30	0.7891	0.5811	0.5278	0.6919
	C INTERMEDIA ADEN+CA	1	0	1	1	0.4170	0.5000	0.7530	0.2704
RECTUM	Fibroma	0	1	0	0	0.5088	0.5000	.	.
SKIN	Adenoma: basal cell	0	0	1	0	0.2616	.	0.5059	.
SK N MISC.	Benign melanoma	0	1	0	0	0.5088	0.5000	.	.
	Carcinoma: squamous	1	0	1	0	0.6388	0.5000	0.7530	0.5172
	Fibroma	0	1	1	0	0.5174	0.5000	0.5000	.
STOMACH	Leiomyosarcoma	0	0	1	0	0.2616	.	0.5059	.
SUBCU. TISSUE	Fibroma	0	0	2	1	0.2170	.	0.2470	0.5172
	Lipoma	0	0	0	1	0.2632	.	.	0.5172
	Osteosarcoma	1	0	0	0	0.7544	0.5000	0.5000	0.5172
SYSTEMIC	Histiocytic sarcoma	1	0	1	0	0.6358	0.4941	0.7529	0.5114
	Leukemia: granulocyt	0	0	1	0	0.2632	.	0.5000	.
	Malignant lymphoma	1	0	1	0	0.6358	0.4941	0.7529	0.5114
THYMUS	Benign thymoma	3	6	5	6	0.3003	0.2414	0.3563	0.2779
	Hemangiosarcoma	0	0	0	1	0.2632	.	.	0.5172
	Malignant thymoma	0	1	0	0	0.5088	0.5000	.	.
THYRO D	Adenoma: C-cell	6	5	4	7	0.3527	0.4830	0.6308	0.5358
	Adenoma: follicular	1	0	1	4	0.0250	0.4941	0.7471	0.1947
	Carcinoma: C-cell	1	0	2	0	0.6607	0.5000	0.5000	0.5172
	C CCELL ADEN+CAR	7	5	6	7	0.4445	0.6046	0.5000	0.4222
UTERUS	Adenocarcinoma: endometrial	0	2	2	1	0.4657	0.2470	0.2529	0.5172
	Adenoma: endometrial	0	0	1	1	0.1981	.	0.5000	0.5172
	Hemangiosarcoma	0	0	1	0	0.2616	.	0.5059	.
	Malignant schwannoma	1	0	2	0	0.6564	0.4941	0.5087	0.5114
	Polyp: endometrial s	1	9	6	5	0.4245	0.0074*	0.0548	0.1120
	Sarcoma: endometrial	0	2	0	0	0.7602	0.2470	.	.
	C ENDOMETRIAL ADEN+C	0	2	3	2	0.2689	0.2470	0.1249	0.2646
	C POLYP SARCOMA ENDO	1	11	6	5	0.5153	0.0017*	0.0548	0.1120
Combined organ	C Hemangioma+Hemangiosarcoma	0	1	1	3	0.0495	0.5000	0.5059	0.1339

Table 6A: Tumor Rates and P-Values for Dose Response Relationship and Pairwise Comparisons – Male Mice

Organ Name	Tumor Name	0	150	400	1200	P-Value
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		mg/kg/day C N=70	mg/kg/day LD N=70	mg/kg/day MD N=70	mg/kg/day HD N=70	Dos Response	C vs. LD	C vs. MD	C vs. HD
ABDOMEN	Hemangiosarcoma	0	0	0	1	0.2523	.	.	0.4828
ADRENAL	Adenoma: subcapsular	0	1	1	2	0.1159	0.4690	0.4690	0.2309
BONE MISC.	Osteosarcoma	1	0	0	0	0.7285	0.4690	0.4690	0.4783
CECUM	Adenocarcinoma	0	0	1	1	0.1838	.	0.4690	0.4828
DUODENUM	Adenocarcinoma	0	0	1	0	0.2489	.	0.4690	.
	Sarcoma (not otherwi	0	0	1	0	0.2477	.	0.4737	.
EAR	Neural crest tumor	1	0	0	0	0.7285	0.4690	0.4690	0.4783
ESOPHAGUS	Papilloma: squamous	0	0	1	0	0.2489	.	0.4690	.
EYE	Carcinoma (not other	0	1	0	0	0.4887	0.4690	.	.
FAT	Carcinoma (not other	0	0	0	1	0.2489	.	.	0.4783
HARDERIAN GL	Adenoma	14	10	12	11	0.6064	0.6542	0.4664	0.6007
LEUM	Adenocarcinoma	1	0	0	0	0.7285	0.4690	0.4690	0.4783
JEJUNUM	Adenocarcinoma	0	0	0	2	0.0645	.	.	0.2352
K DNEY	Adenoma: tubular cel	1	0	2	1	0.3664	0.4690	0.4532	0.7346
	Carcinoma: tubular c	0	1	1	2	0.1157	0.4690	0.4737	0.2309
	Hemangiosarcoma	0	0	0	1	0.2489	.	.	0.4783
	C TUBULAR ADEN+CAR	1	1	3	3	0.1416	0.7203	0.2704	0.2836
LARYNX	Papilloma: squamous	0	0	1	0	0.2489	.	0.4690	.
LIVER	Adenoma: hepatocellu	14	11	18	12	0.5484	0.5420	0.1823	0.4894
	Carcinoma: hepatocel	3	5	4	0	0.9699	0.2911	0.4410	0.8614
	Hemangioma	0	0	1	0	0.2489	.	0.4690	.
	Hemangiosarcoma	4	4	1	3	0.6212	0.5811	0.7770	0.4613
	C HEPATOCELLULAR ADE	15	15	19	12	0.7266	0.4527	0.1992	0.5729
LUNG	Adenoma: alveolar/br	18	9	14	8	0.9461	0.9125	0.6546	0.9614
	Carcinoma: alveolar/	16	12	17	6	0.9801	0.5933	0.3224	0.9672
	Sarcoma (not otherwi	0	1	0	0	0.4887	0.4690	.	.
	C ALVEOLAR ADEN+CAR	28	19	29	13	0.9890	0.8191	0.3375	0.9878
PITUITARY	Adenoma: pars interm	1	1	1	0	0.7202	0.7203	0.7203	0.4783
PREPUTIAL GLAND	Carcinoma	0	1	0	0	0.4887	0.4690	.	.
PROSTATE	Adenocarcinoma	1	0	0	0	0.7285	0.4690	0.4690	0.4783
	Adenoma	0	1	0	0	0.4887	0.4690	.	.
	C ADEN+CAR	1	1	0	0	0.7964	0.7203	0.4690	0.4783
SAL. GL. PAROTI	Sarcoma (not otherwi	1	0	0	0	0.7285	0.4690	0.4690	0.4783
SALIV.GL. MANDI	Malignant myoepithel	1	0	0	0	0.7285	0.4690	0.4690	0.4783
SEM NAL VESICLE	Sarcoma (not otherwi	0	0	1	0	0.2477	.	0.4737	.
SK N MISC.	Carcinoma: squamous	1	0	0	1	0.4367	0.4690	0.4690	0.7300
	Papilloma: squamous	0	0	1	0	0.2489	.	0.4690	.
	C SQUAMOUSE ADEN+CAR	1	0	1	1	0.3820	0.4690	0.7203	0.7300
SP NAL CORD CER	Malignant astrocytom	1	0	0	0	0.7285	0.4690	0.4690	0.4783
SPLEEN	Hemangiosarcoma	0	3	0	1	0.4981	0.1001	.	0.4783
STOMACH	Adenocarcinoma	1	0	1	0	0.6089	0.4690	0.7203	0.4783
	Carcinoma: squamous	0	1	0	1	0.3048	0.4690	.	0.4828
	Sarcoma (not otherwi	1	0	0	0	0.7285	0.4690	0.4690	0.4783
	C SEQUAMOUSE ADEN+CAR	1	1	1	1	0.5038	0.7203	0.7203	
SUBCU. TISSUE	Fibrosarcoma	1	0	1	2	0.1656	0.4690	0.7203	0.4739
	Lipoma	0	0	0	1	0.2523	.	.	0.4828
	Malignant histiocyti	0	0	1	0	0.2489	.	0.4690	.
SYSTEMIC	Histiocytic sarcoma	2	3	5	4	0.2462	0.4499	0.1918	0.3073
	Malignant lymphoma	11	4	10	6	0.7187	0.9227	0.4327	0.7941
	Mast cell tumor	1	0	0	1	0.4417	0.4690	0.4690	0.7346
TESTIS	Adenoma: interstitia	2	1	3	3	0.2244	0.4532	0.4410	0.4586

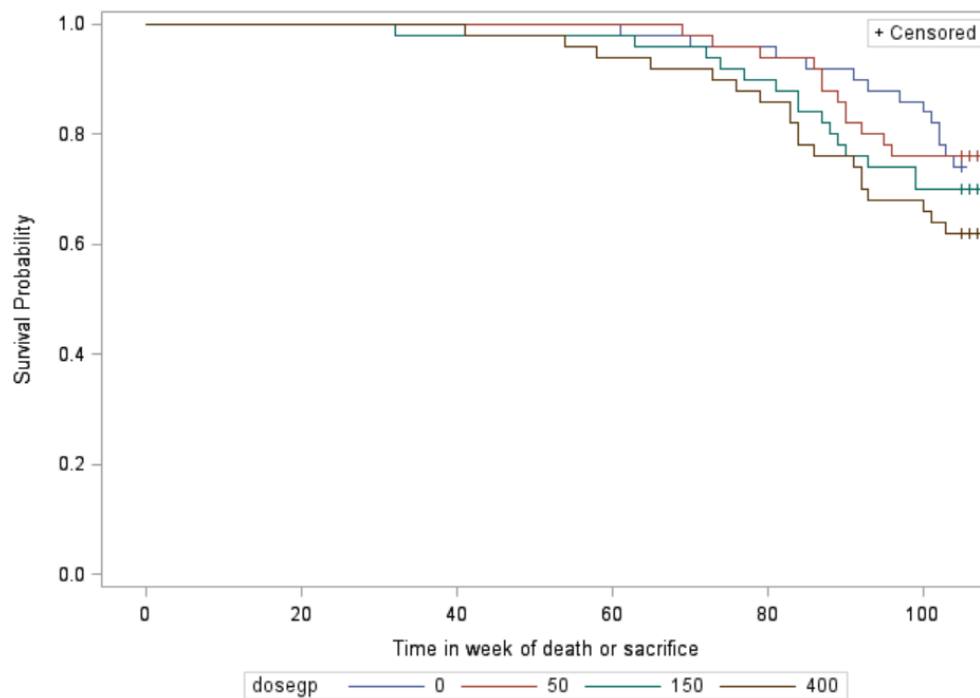
Organ Name	Tumor Name	0	150	400	1200	P-Value			
		mg/kg/day C N=70	mg/kg/day LD N=70	mg/kg/day MD N=70	mg/kg/day HD N=70	Dos Response	C vs. LD	C vs. MD	C vs. HD
THORAX	Hemangioma	0	0	0	1	0.2489	.	.	0.4783
	Mesothelioma(M)	0	0	0	1	0.2489	.	.	0.4783
THYRO D	Adenoma: follicular	0	0	0	2	0.0611	.	.	0.2265
	Carcinoma: follicula	1	0	1	0	0.6089	0.4690	0.7203	0.4783
	C FOLLICULAR ADEN+CA	1	0	1	2	0.1617	0.4690	0.7203	
TONGUE	Papilloma: squamous	0	0	0	1	0.2489	.	.	0.4783
URINARY BLADDER	Hemangiosarcoma	1	0	0	0	0.7285	0.4690	0.4690	0.4783
	Leiomyosarcoma	0	0	0	1	0.2489	.	.	0.4783

Table 6B: Tumor Rates and P-Values for Dose Response Relationship and Pairwise Comparisons – Female Mice

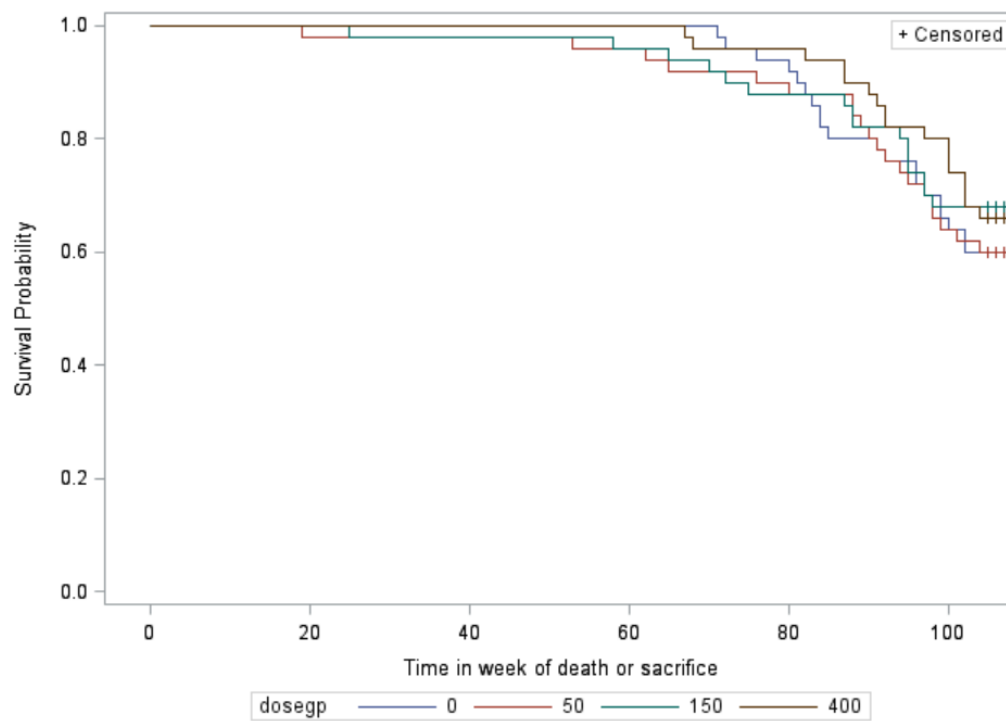
Organ Name	Tumor Name	0	150	400	1200	P-Value			
		mg/kg/day C N=70	mg/kg/day LD N=70	mg/kg/day MD N=70	mg/kg/day HD N=70	Dos Response	C vs. LD	C vs. MD	C vs. HD
ABDOMEN	Hemangiosarcoma	0	1	0	0	0.5146	0.4700	.	.
ADRENAL	Adenoma: cortical	1	0	0	0	0.7415	0.4646	0.4952	0.5047
	Adenoma: spindle cel	1	0	1	1	0.4173	0.4646	0.7476	0.2523
	Adenoma: subcapsular	0	1	0	0	0.5171	0.4646	.	.
	Benign pheochromocyt	1	0	0	1	0.4637	0.4646	0.4952	0.2570
BONE MISC.	Osteosarcoma	0	1	0	2	0.1189	0.4700	.	0.2570
CLITORAL GLAND	Adenocarcinoma	0	1	0	0	0.5146	0.4700	.	.
HARDERIAN GL	Adenocarcinoma	1	1	0	1	0.5485	0.7216	0.4952	0.2523
	Adenoma	2	4	4	3	0.4874	0.2834	0.3300	0.5089
	C ADEN+CAR	3	5	4	4	0.4818	0.2827	0.4788	0.5000
HEART	Sarcoma (not otherwi	0	0	1	0	0.2634	.	0.4952	.
LIVER	Adenoma: hepatocellu	2	1	0	1	0.6450	0.4546	0.7476	0.5071
	Hemangiosarcoma	0	0	1	0	0.2634	.	0.4952	.
LUNG	Adenoma: alveolar/br	11	2	2	3	0.9686	0.9851	0.9925	0.9809
	Carcinoma: alveolar/	5	3	9	5	0.4798	0.5729	0.2168	0.3948
	C ALVEOLAR ADEN+CAR	16	5	11	8	0.8888	0.9835	0.8270	0.9530
MAMMARY GLAND	Adenocarcinoma	4	2	0	0	0.9949	0.5926	0.9363	0.9410
MESENTERIC LN	Hemangioma	0	0	1	0	0.2634	.	0.4952	.
MUSCLE MISC.	Rhabdomyosarcoma	0	0	1	0	0.2634	.	0.4952	.
OVARY	Benign luteoma	2	1	0	0	0.9386	0.4465	0.7476	0.7570
	Benign thecoma	0	0	1	0	0.2634	.	0.4952	.
	Cystadenoma	2	1	4	2	0.4728	0.4465	0.3482	0.3160
	Hemangiosarcoma	1	0	0	0	0.7415	0.4646	0.4952	0.5047
	Leiomyoma	0	1	0	0	0.5171	0.4646	.	.
	Malignant granulosa-	0	1	0	0	0.5171	0.4646	.	.
	Malignant luteoma	0	0	1	0	0.2621	.	0.5000	.
	C LUTEOMA Benign+Mali	2	1	1	0	0.8874	0.4465	0.5000	0.7570
	Adenoma	1	0	0	0	0.7415	0.4646	0.4952	0.5047
PARATHYROID GL.	Adenoma: pars distal	4	3	0	1	0.9327	0.4443	0.9387	0.8243
	Adenoma: pars interm	1	1	0	2	0.2601	0.7216	0.4952	0.5140
	Carcinoma: pars inte	0	0	0	1	0.2634	.	.	0.5047
	C INTERMEDIA ADEN+CA	1	1	0	3	0.1062	0.7216	0.4952	0.3231
SK N MISC.	Carcinoma: basal cel	0	1	0	0	0.5146	0.4700	.	.
	Carcinoma: squamous	1	1	0	0	0.8175	0.7159	0.4952	0.5047

Organ Name	Tumor Name	0	150	400	1200	P-Value			
		mg/kg/day C N=70	mg/kg/day LD N=70	mg/kg/day MD N=70	mg/kg/day HD N=70	Dos Response	C vs. LD	C vs. MD	C vs. HD
SPLEEN	Hemangiosarcoma	1	2	0	0	0.8999	0.4465	0.4952	0.5047
STOMACH	Adenoma	1	0	0	0	0.7415	0.4646	0.4952	0.5047
	Carcinoma: squamous	0	0	1	0	0.2634	.	0.4952	.
	C SQUAMOUS ADEN+CAR	1	0	1	0	0.6362	0.4646	0.7476	0.5047
SUBCU. TISSUE	Fibrosarcoma	2	2	0	0	0.9604	0.6445	0.7476	0.7570
	Hemangiosarcoma	1	1	0	1	0.4621	0.7159	0.4952	0.2570
SYSTEMIC	Histiocytic sarcoma	6	8	2	5	0.7334	0.3157	0.8525	0.5000
	Malignant lymphoma	16	19	21	20	0.3306	0.2716	0.2310	0.2943
THORAX	Carcinoma (not other	0	0	0	1	0.2634	.	.	0.5047
THYROID	Adenoma: follicular	0	1	1	0	0.5218	0.4646	0.5000	.
UTERUS	Adenocarcinoma: endo	0	0	1	0	0.2634	.	0.4952	.
	Deciduoma	1	0	0	1	0.4584	0.4646	0.4952	0.2523
	Hemangioma	0	0	1	0	0.2634	.	0.4952	.
	Hemangiosarcoma	2	0	1	0	0.8505	0.7159	0.5000	0.7570
	Leiomyoma	3	2	1	4	0.2693	0.4326	0.6839	0.5208
	Leiomyosarcoma	1	0	2	2	0.1919	0.4646	0.4928	0.5071
	Polyp: endometrial s	3	6	5	4	0.5764	0.1873	0.3578	0.5208
	Sarcoma: endometrial	0	2	0	2	0.2072	0.2134	.	0.2523
	C ENDOMETRIAL ADEN+C	3	6	5	6	0.3006	0.1873	0.3578	0.2632
VAGINA	Polyp	1	0	1	0	0.6362	0.4646	0.7476	0.5047

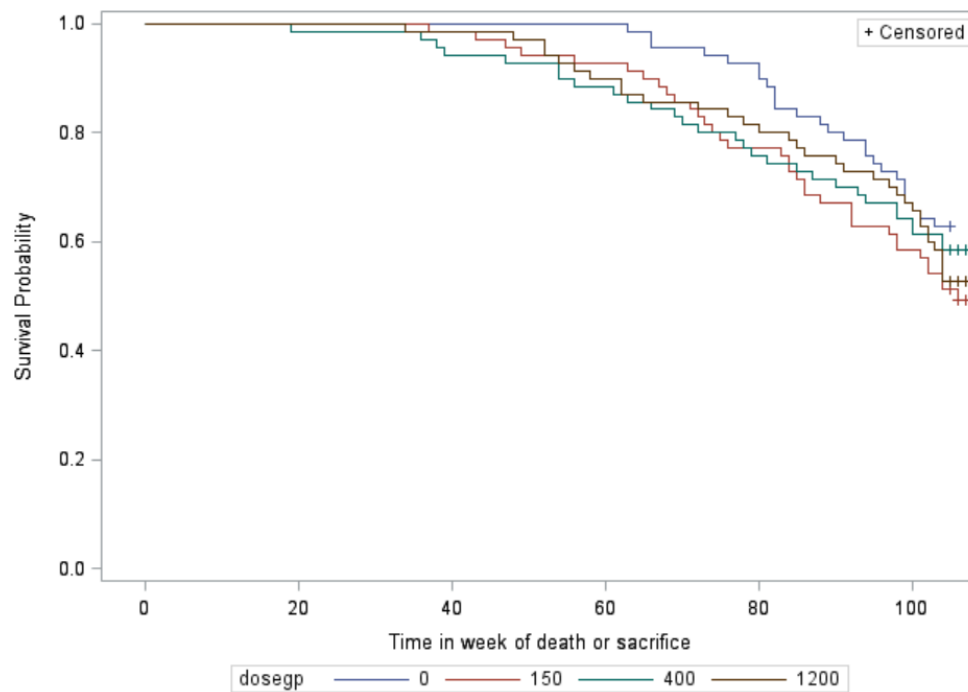
Figure 1A: Kaplan-Meier Survival Functions for Male Rats



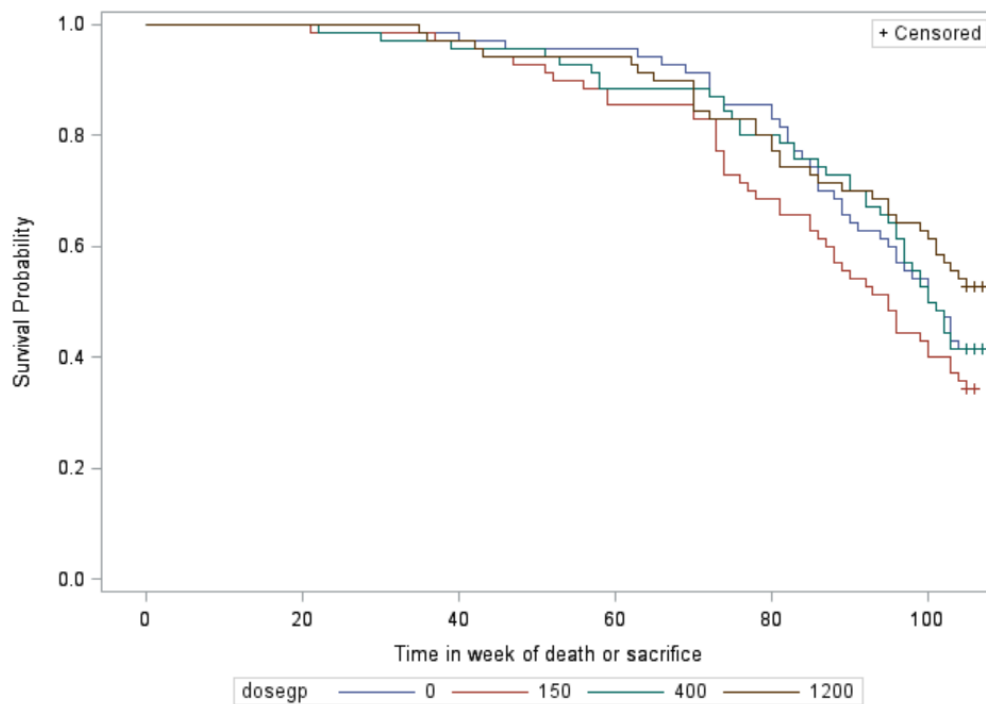
Note: dose group should be 0, 50, 150, or 400-mg/kg/day

Figure 1B: Kaplan-Meier Survival Functions for Female Rats

Note: dose group should be 0, 50, 150, or 400-mg/kg/day

Figure 2A: Kaplan-Meier Survival Functions for Male Mice

Note: dose group should be 0, 150, 40, or 1200-mg/kg/day

Figure 2B: Kaplan-Meier Survival Functions for Female Mice

Note: dose group should be 0, 150, 40, or 1200-mg/kg/day

7 References

1. Peto, R., M.C. Pike, N.E. Day, R.G. Gray, P.N. Lee, S. Parish, J. Peto, Richards, and J. Wahrendorf, "Guidelines for sample sensitive significance test for carcinogenic effects in long-term animal experiments", Long term and short term screening assays for carcinogens: A critical appraisal, International agency for research against cancer monographs, *Annex to supplement, World Health Organization, Geneva*, 311-426, 1980.
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7. Guidance for Industry. Statistical Aspects of the Design, Analysis, and Interpretation of Chronic Rodent Carcinogenicity Studies of Pharmaceuticals (Draft Guidance). U.S. Department of Health and Human Services, Food and Drug Administration, Center for Drug Evaluation and Research (CDER), May 2001.

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/s/

FENG ZHOU
05/18/2015

KARL K LIN
05/18/2015
Concur with review

STATISTICS FILING CHECKLIST FOR A NEW NDA/BLA

NDA Number: 207620

Applicant: Novartis

Stamp Date: 12/17/2014

Drug Name: LCZ696

NDA/BLA Type: NDA

On **initial** overview of the NDA/BLA application for RTF:

	Content Parameter	Yes	No	NA	Comments
1	Index is sufficient to locate necessary reports, tables, data, etc.	X			
2	ISS, ISE, and complete study reports are available (including original protocols, subsequent amendments, etc.)	X			
3	Safety and efficacy were investigated for gender, racial, and geriatric subgroups investigated (if applicable).	X			
4	Data sets in EDR are accessible and do they conform to applicable guidances (e.g., existence of define.pdf file for data sets).	X			

IS THE STATISTICAL SECTION OF THE APPLICATION FILEABLE? __Yes__

If the NDA/BLA is not fileable from the statistical perspective, state the reasons and provide comments to be sent to the Applicant.

Please identify and list any potential review issues to be forwarded to the Applicant for the 74-day letter.

Content Parameter (possible review concerns for 74-day letter)	Yes	No	NA	Comment
Designs utilized are appropriate for the indications requested.	X			
Endpoints and methods of analysis are specified in the protocols/statistical analysis plans.	X			
Interim analyses (if present) were pre-specified in the protocol and appropriate adjustments in significance level made. DSMB meeting minutes and data are available.	X			
Appropriate references for novel statistical methodology (if present) are included.	X			
Safety data organized to permit analyses across clinical trials in the NDA/BLA.	X			
Investigation of effect of dropouts on statistical analyses as described by applicant appears adequate.	X			

File name: 5_Statistics Filing Checklist for a New NDA_BLA110207

STATISTICS FILING CHECKLIST FOR A NEW NDA/BLA

John Lawrence	2/2/2015
Reviewing Statistician	Date
Supervisor/Team Leader	Date

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/s/

JOHN P LAWRENCE
02/02/2015

HSIEN MING J HUNG
02/03/2015